



Weatherford®

COMPENSATED SONIC

COMPANY		S.N.G.N. ROMGAZ S.A.			
WELL		209 NADES			
FIELD		NADES			
PROVINCE/COUNTY		MURES			
COUNTRY/STATE		ROMANIA			
LOCATION		X = 529925.187 m Y = 477271.675 m			
FIELD PRINT					
altitude	46°16'07.03" N	Other Services		PHOTODENSITY	
Longitude	24°42'12.69" E	DUAL LATEROLOG MICROLATEROLOG		DUAL NEUTRON	
Permanent Datum GL, Elevation 480.21 metres		BOREHOLE NAVIGATION		CBL	
Log Measured From KB, 4.50 metres above Permanent Datum				Elevations: KB 484.71 DF 484.71 GL 480.21	
Drilling Measured From KB					
Date	04-SEP-2024				
Run Number	2A				
Service Order	24220368				
Depth Driller	1225.00	metres			
Depth Logger	1226.00	metres			
First Reading	1212.00	metres			
Last Reading	250.00	metres			
Casing Driller	250.00	metres			
Casing Logger	249.50	metres			
Bit Size	8.500	inches			
Hole Fluid Type	INHIBITIV				
Density / Viscosity	1.10 g/cc	50.00 SEC/L			
PH / Fluid Loss	9.50	4.10 ml/30Min			
Sample Source	FLOWLINE				
Rm @ Measured Temp	0.15 @ 25.0	ohm-m			
Rmf @ Measured Temp	0.12 @ 25.0	ohm-m			
Rmc @ Measured Temp	0.18 @ 25.0	ohm-m			
Source Rmf / Rmc	PRESS	FILTER			
Rm @ BHT	0.11 @ 39.0	ohm-m			
Time Since Circulation	6 HOURS				
Max Recorded Temp	39.00	deg C			
Equipment / Base	13347	PLO			
Recorded By	MALEA CRISTIAN				
Witnessed By	VASILE FARCAS				

In interpreting, communicating or providing information and/or making recommendations, either written or oral, as to logs or test or other data, type or amount of material, or Work or other service to be furnished, or manner of performance, or in predicting results to be obtained, the Contractor will give the Company the benefit of the Contractor's best judgment based on its experience and will perform all such Work in a good and workmanlike manner. Any interpretation of test or other data, and any recommendation or reservoir description based upon such interpretations, are opinions based upon inferences from measurements and empirical relationships and assumptions, which inferences and assumptions are not infallible, and with respect to which professional engineers and analysts may differ. ACCORDINGLY ANY INTERPRETATION OR RECOMMENDATION RESULTING FROM THE SERVICES WILL BE AT THE SOLE RISK OF THE COMPANY, AND THE CONTRACTOR CANNOT AND DOES NOT WARRANT THE ACCURACY, CORRECTNESS OR COMPLETENESS OF ANY SUCH INTERPRETATION OR RECOMMENDATION, WHICH INTERPRETATIONS AND RECOMMENDATIONS SHOULD NOT, THEREFORE, UNDER ANY CIRCUMSTANCES BE RELIED UPON AS THE SOLE OR MAIN BASIS FOR ANY DRILLING, COMPLETION, WELL TREATMENT, PRODUCTION OR FINANCIAL DECISION, OR ANY PROCEDURE INVOLVING ANY RISK TO THE SAFETY OF ANY DRILLING ACTIVITY, DRILLING RIG OR ITS CREW OR ANY OTHER INDIVIDUAL. THE COMPANY HAS FULL RESPONSIBILITY FOR ALL DECISIONS CONCERNING THE SERVICES.

BOREHOLE RECORD				Last Edited: 04-SEP-2024 10:04
Bit Size inches	Depth From metres		Depth To metres	
12.250	23.00		246.00	
8.500	250.00		1225.00	
CASING RECORD				
Type	Size inches	Depth From metres	Shoe Depth metres	Weight pounds/ft
SURFACE	16.000	0.00	23.00	0.00
CASING	9.625	0.00	250.00	43.50

REMARKS

Clients intervals and scales logged

Acquisition software: Weatherford Well Manager, version 24.03.1279

Cable used: CAMESA size 15/32", type 7J46RXZ with cable head and 8k weakpoint set

Depth wheel calibration factor used 0.6096

Depth Control Procedure for Multiple runs in the well followed.

Run2A: CBHC, MTAK, MBE, MBE, SHA, MCG, MBN, MTC-Y, MSS, MTC-X, SKJ, MUG, MLE, MLG, MTI, MGS,HFS

Run2B: CBHC, MTAK, SHA, MCG, SKJ, MMR, MDN, MPD, HFS

Run2C: CBHC, MTAK, SHA, MCG, MISD, SKJ, MISE, SKJ, MISD, MSS, MISD, HFS logged in combination.

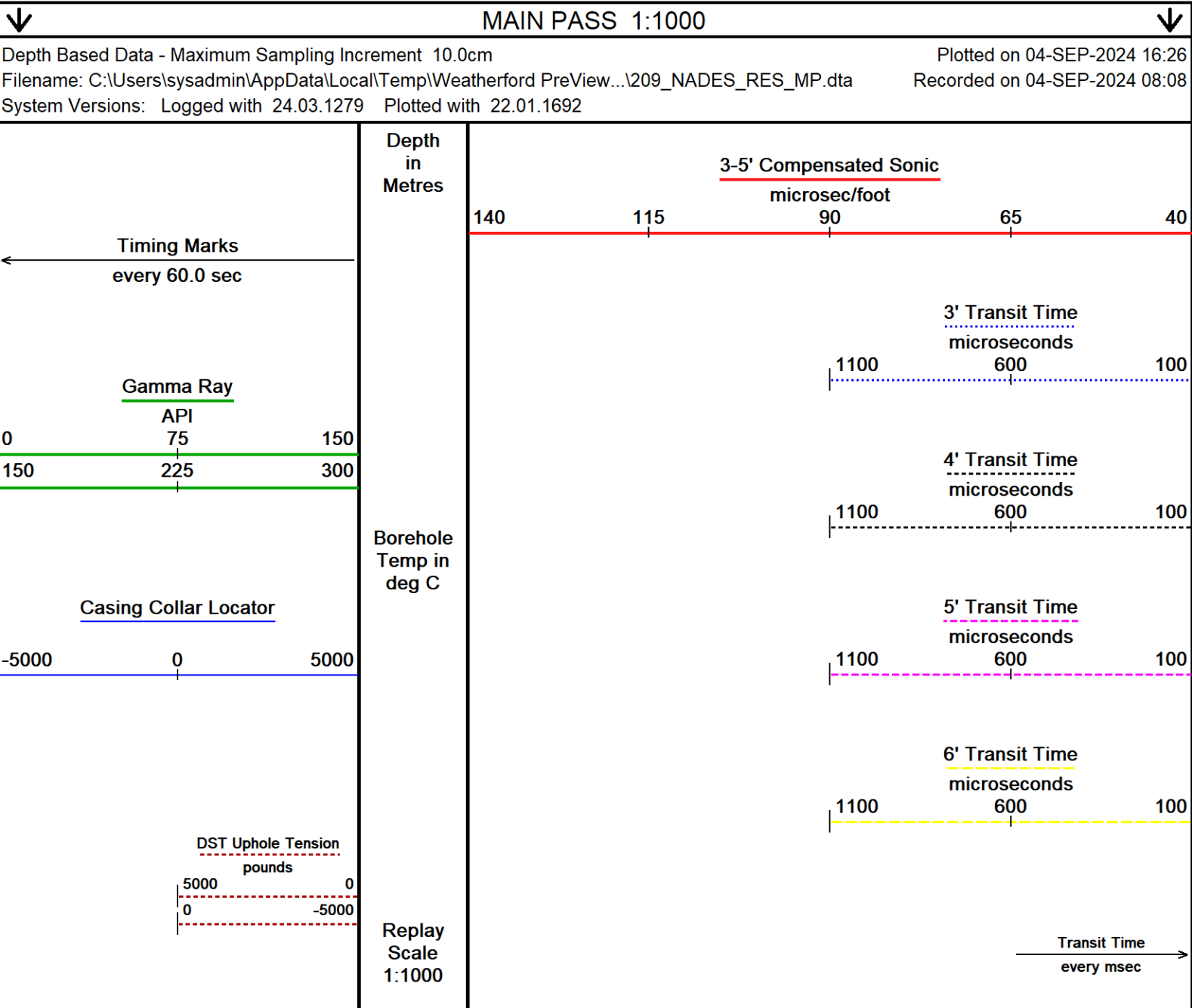
Rm, Rmf, Rmc measured on samples from Mud Engineer given on 04.09.2024

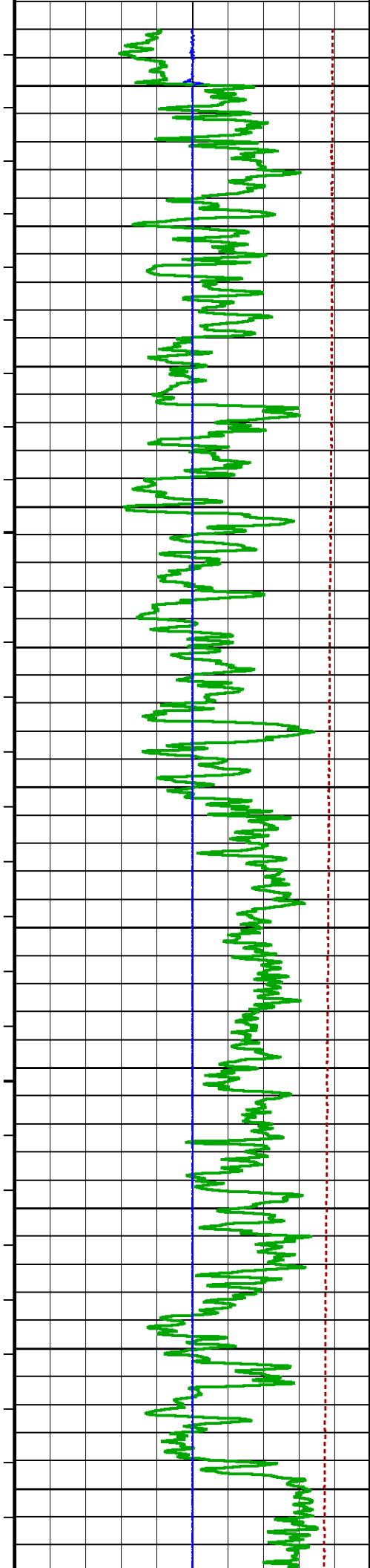
Caliper measurements and averages from TD to casing:

Total hole volume = 35.32 cubic meters

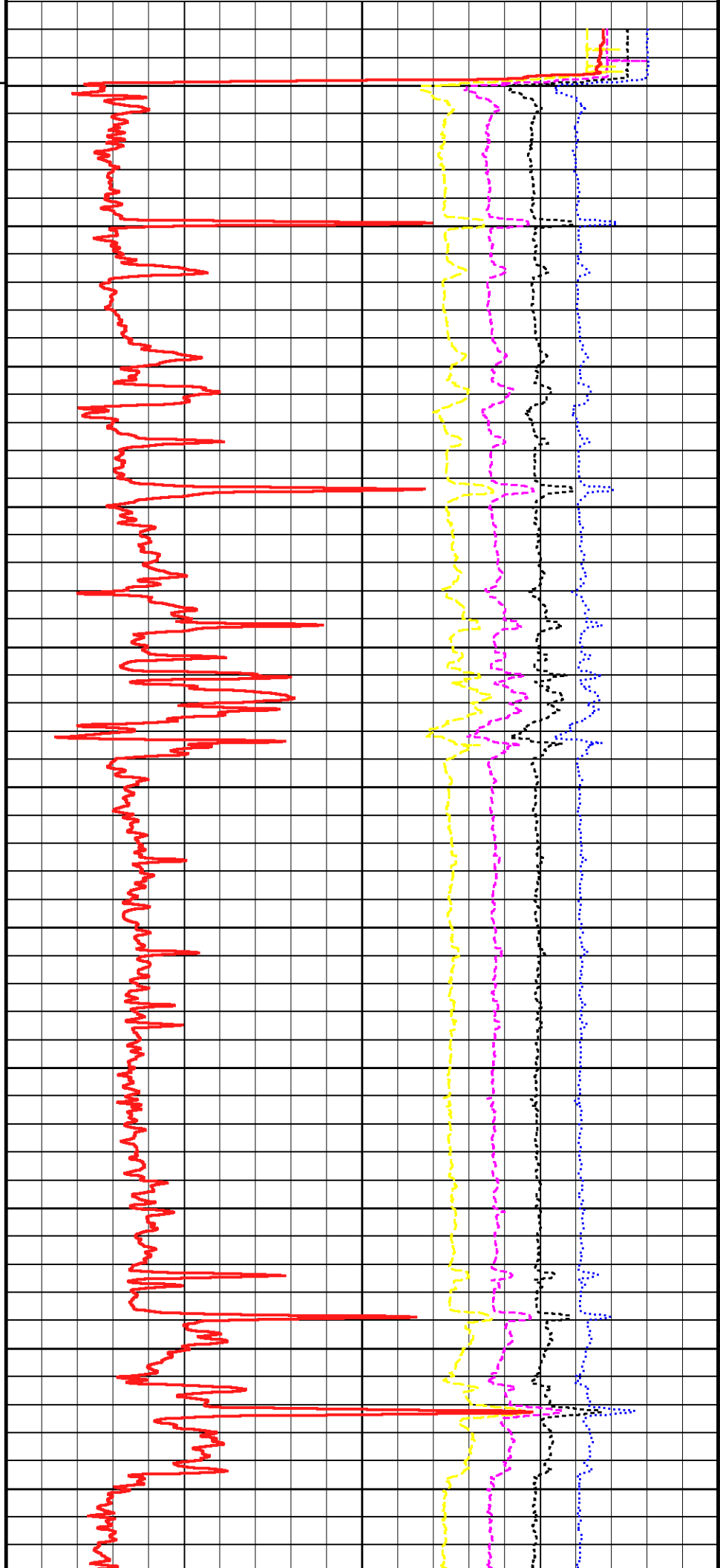
Annular volume = 20.52 cubic meters, based on future casing size of 5.5 inches

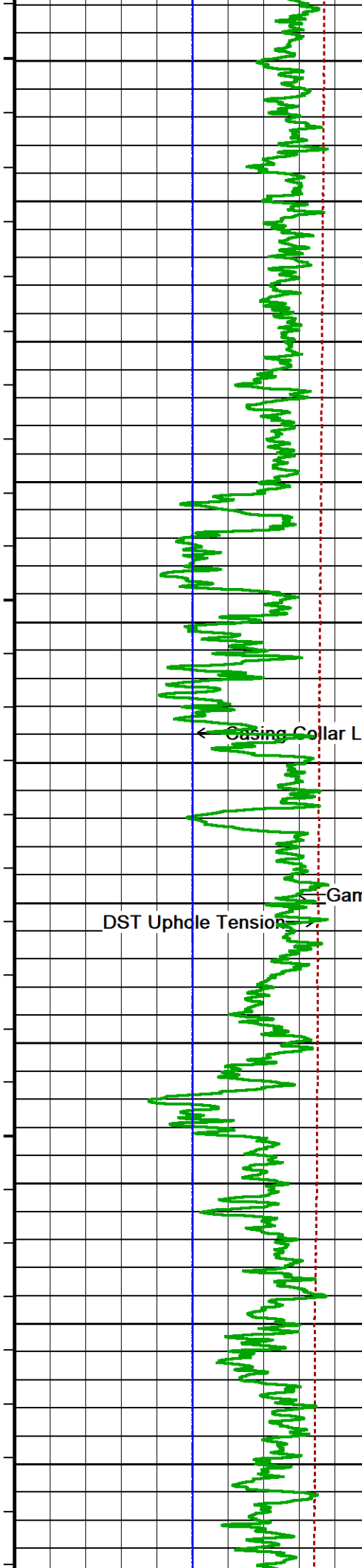
Average hole diameter = 8.49 inches (215.77 mm), based on XY caliper





236
Casing
Shoe
250
31°
300
31°
350
31°
400
32°
450
32°
500





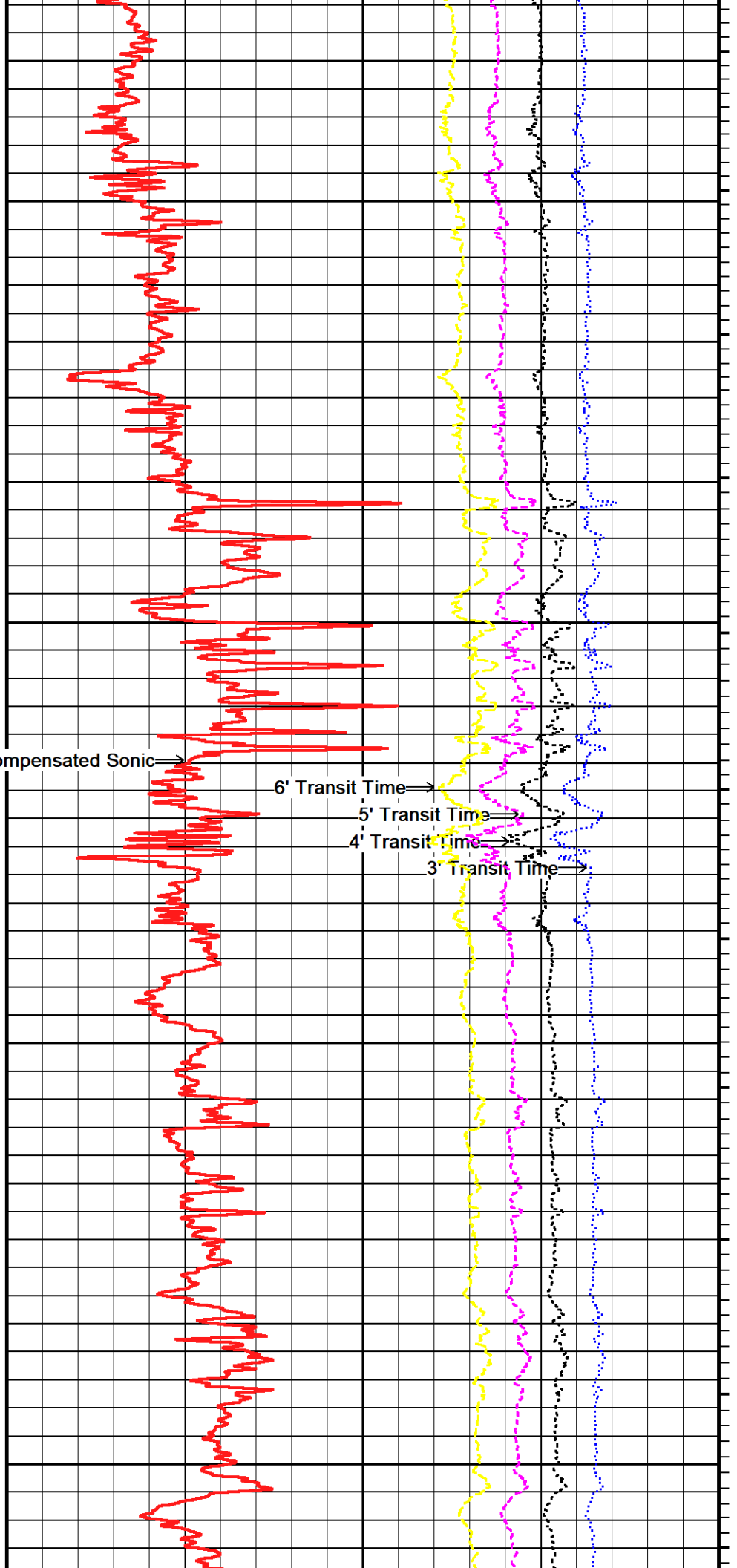
33° 550 33° 600 34° 650 34° 700 35° 750 35°

← Casing Collar Locator

650 Compensated Sonic →

← Gamma Ray

DST Uphole Tension

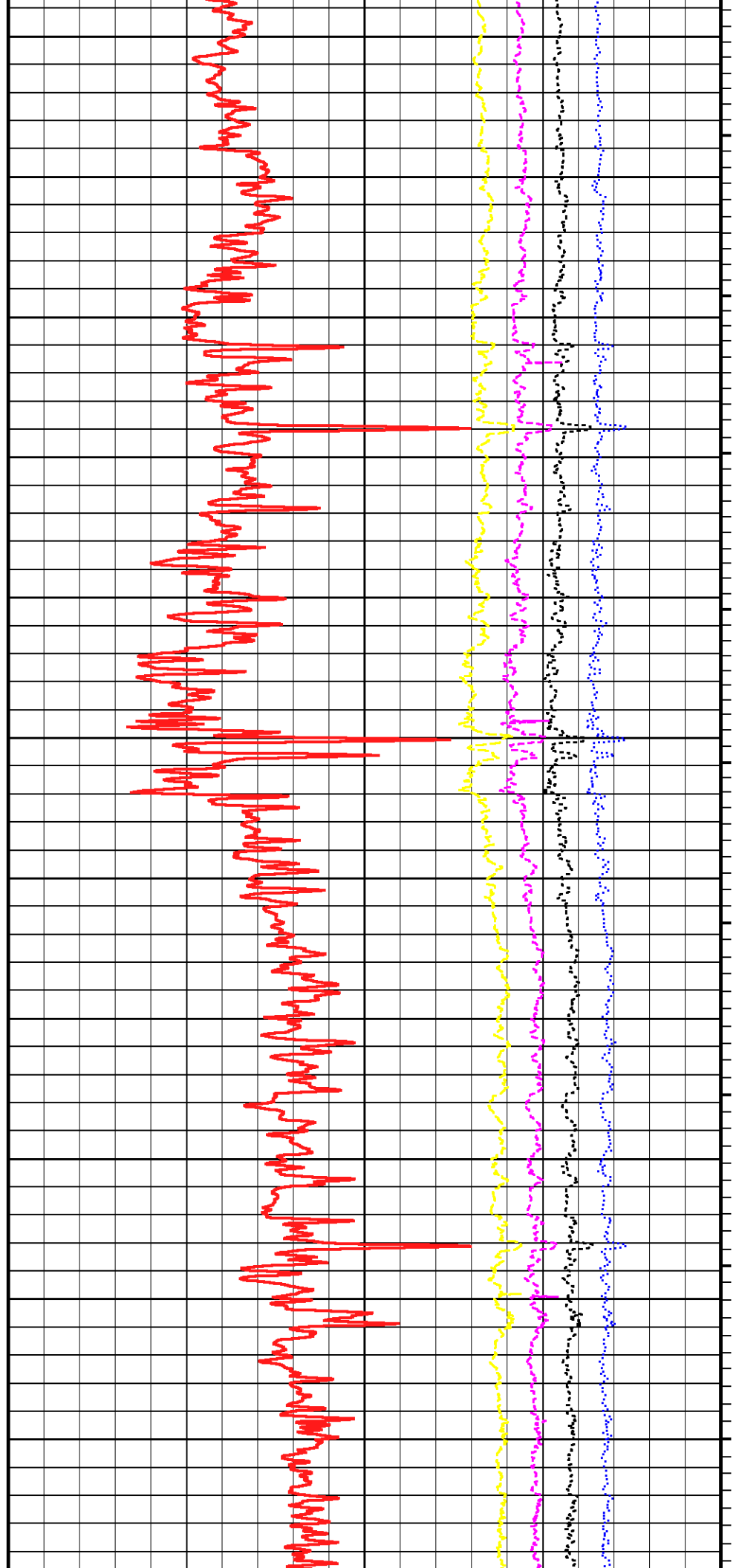
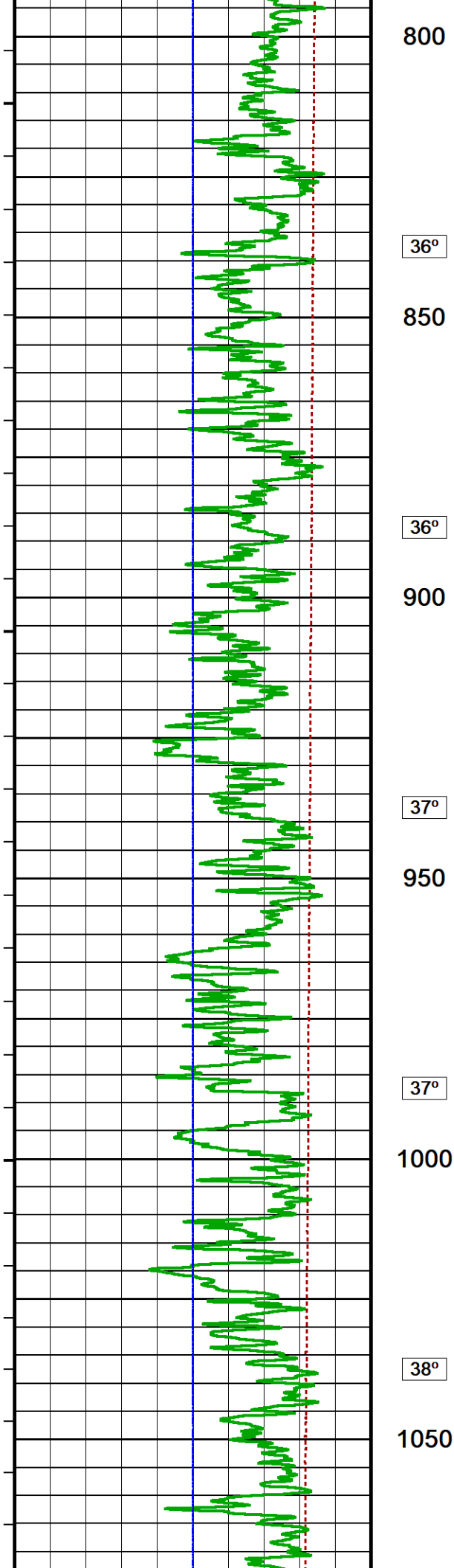


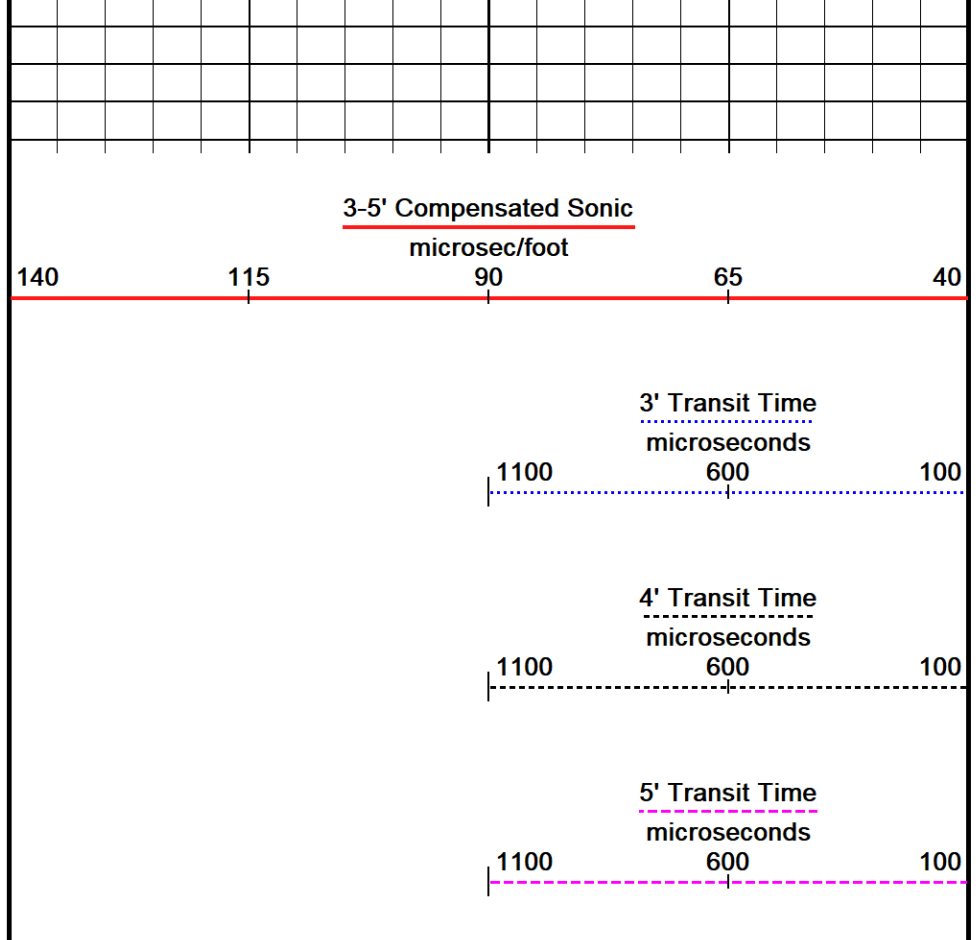
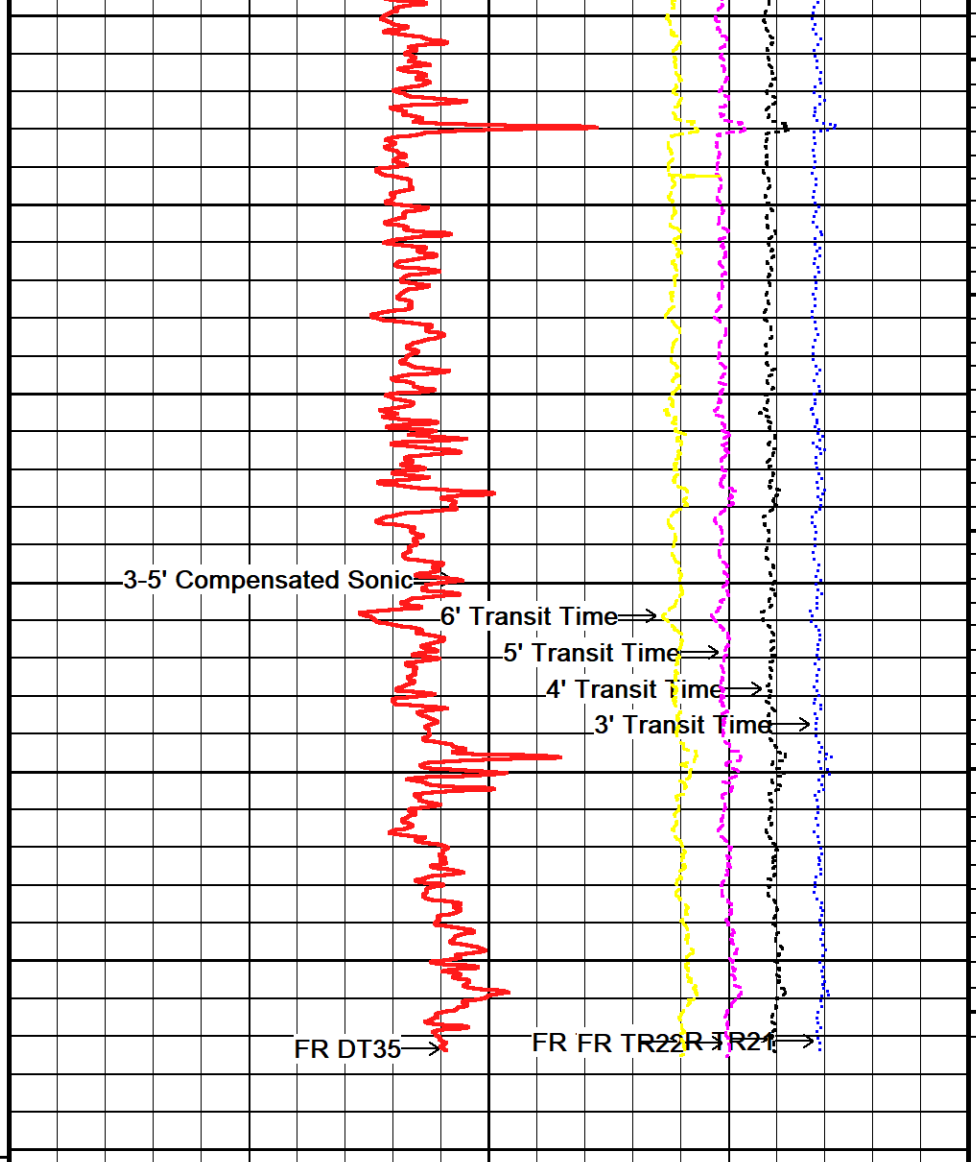
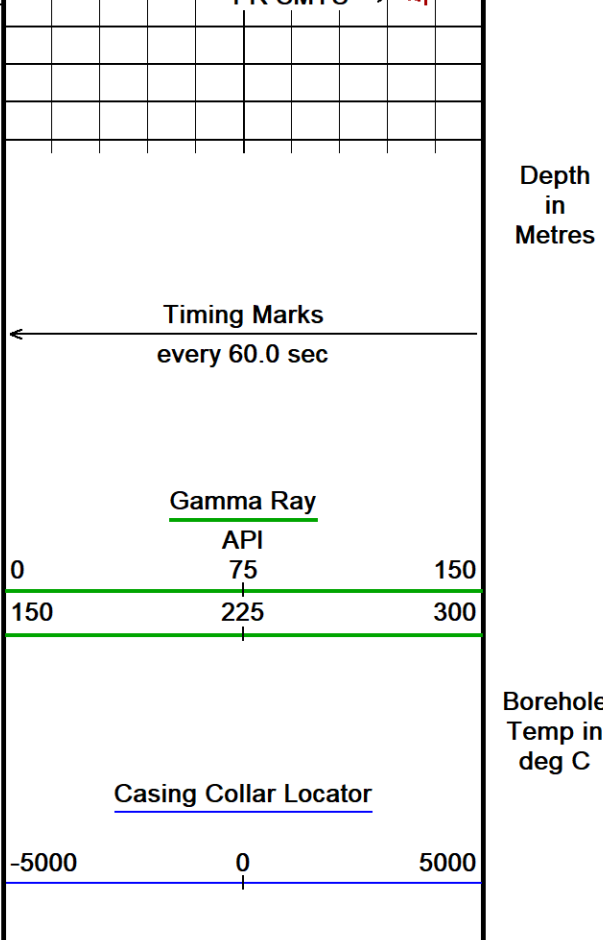
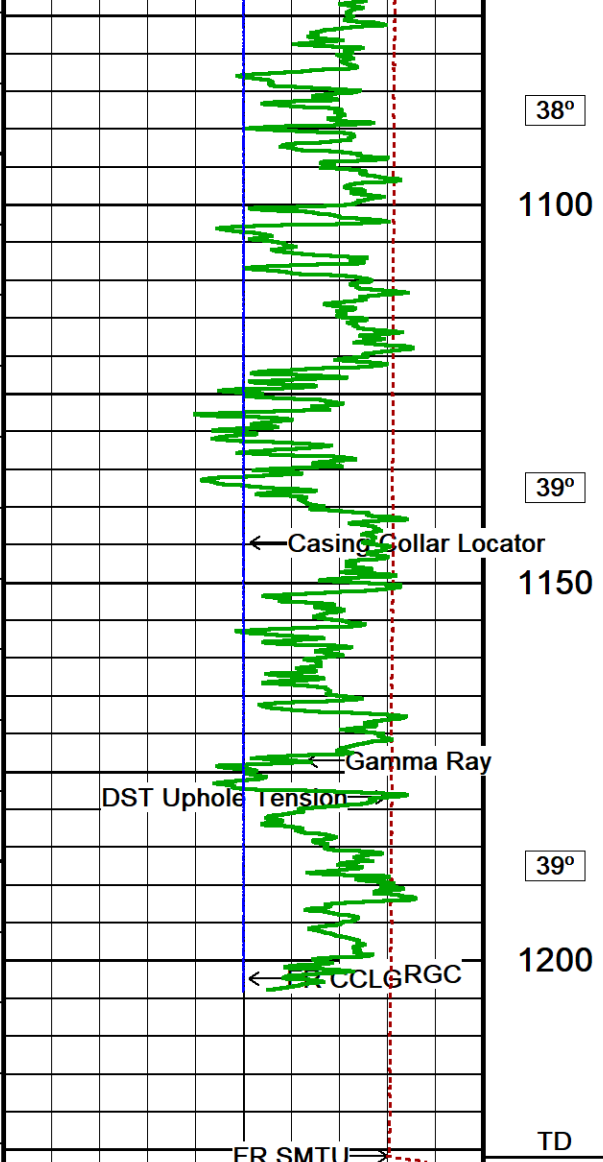
6' Transit Time →

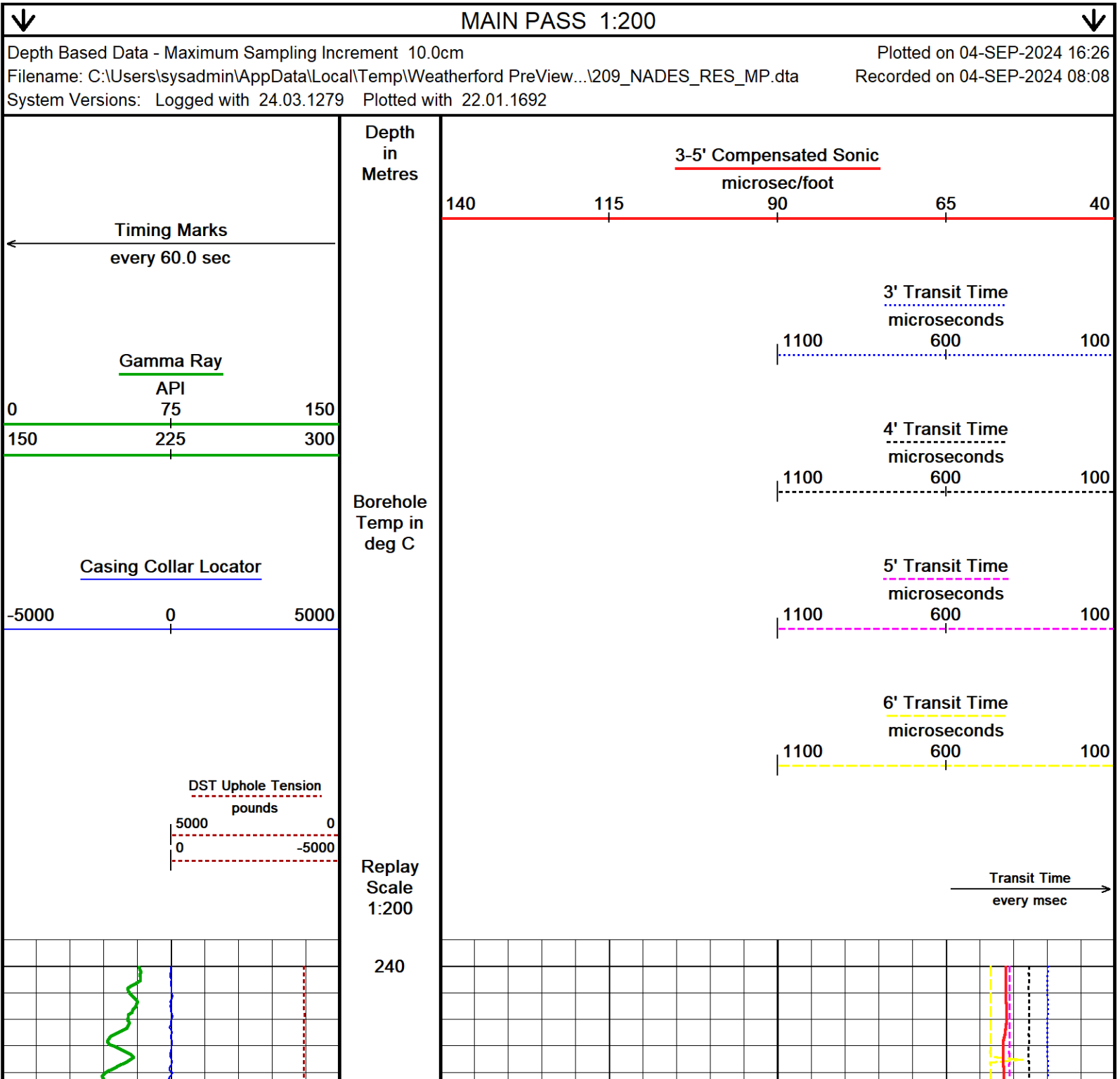
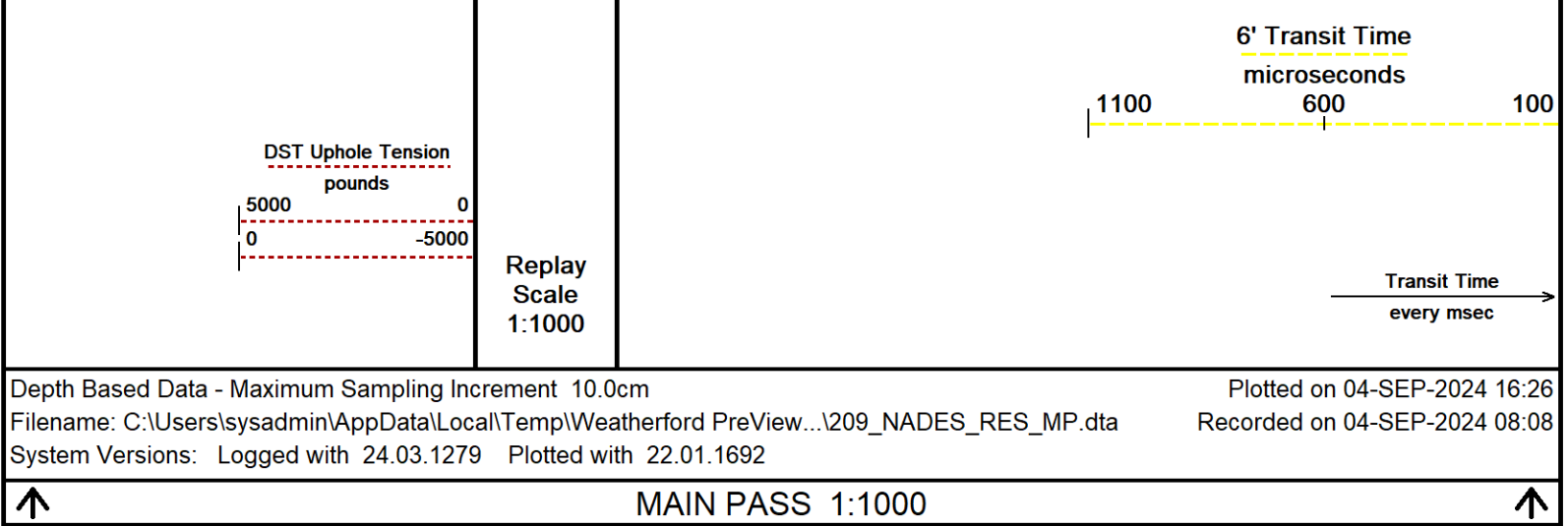
5' Transit Time →

4' Transit Time →

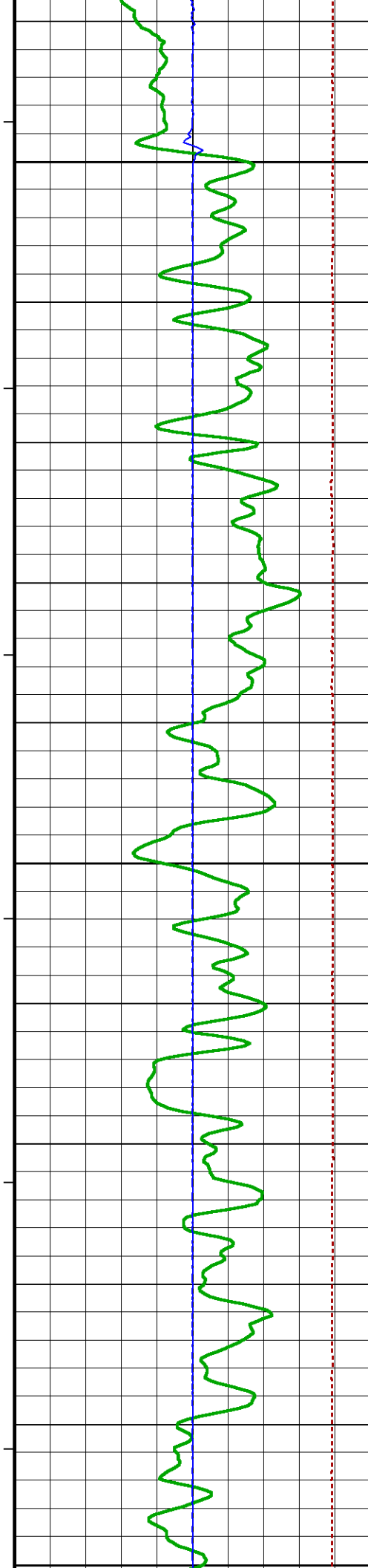
3' Transit Time →





Depth Based Data - Maximum Sampling Increment 10.0cmPlotted on 04-SEP-2024 16:26
Filename: C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView...\209_NADES_RES_MP.dtaRecorded on 04-SEP-2024 08:08
System Versions: Logged with 24.03.1279 Plotted with 22.01.1692

↓MAIN PASS 1:200↓



30°
Casing
Shoe
250

30°

260

30°

270

30°

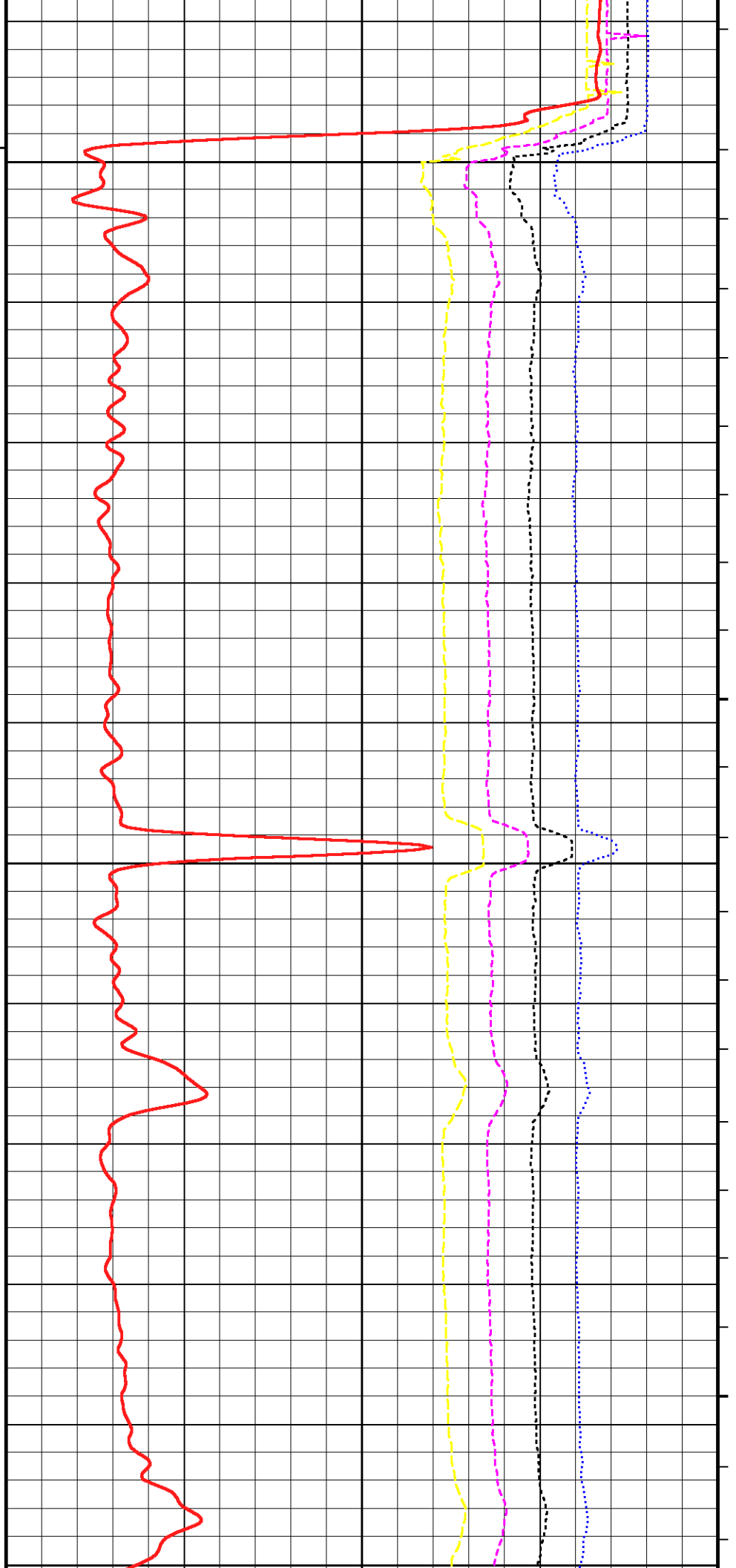
280

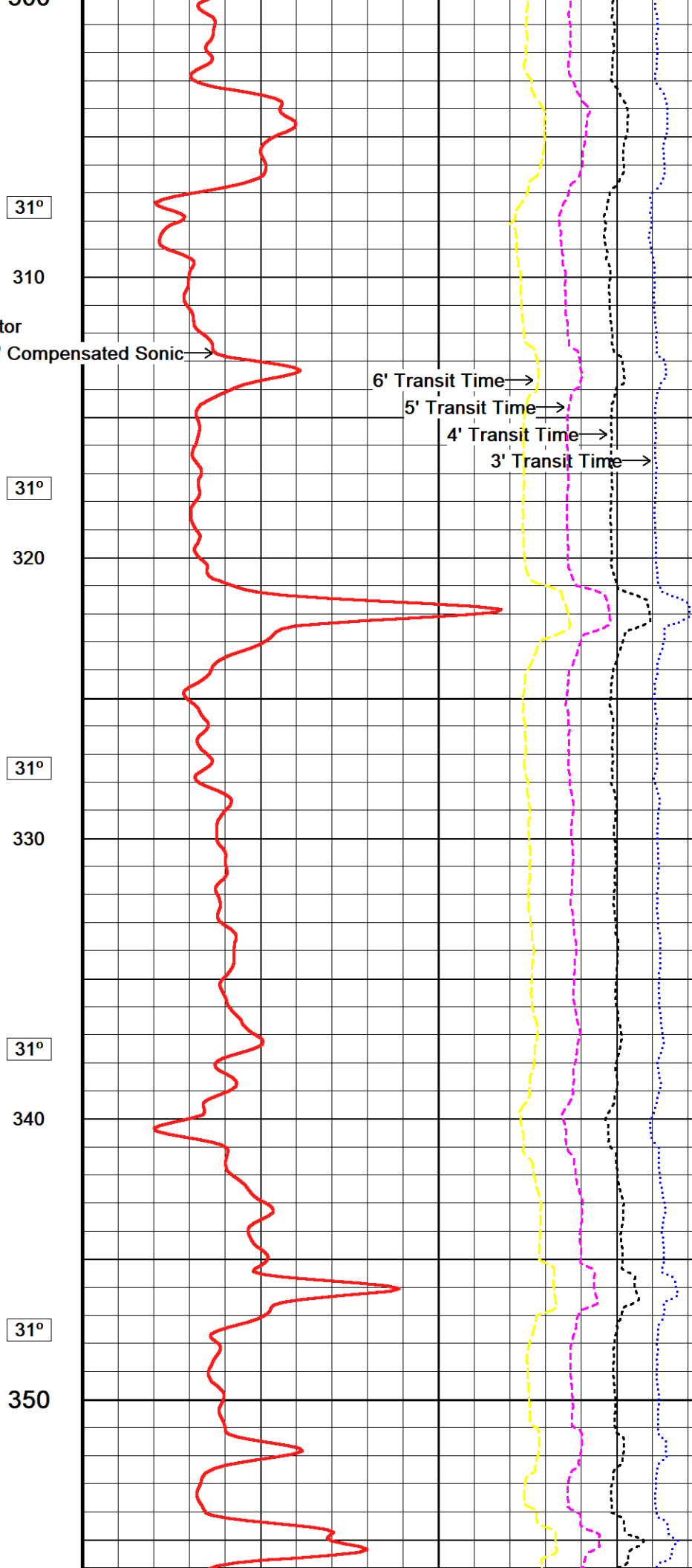
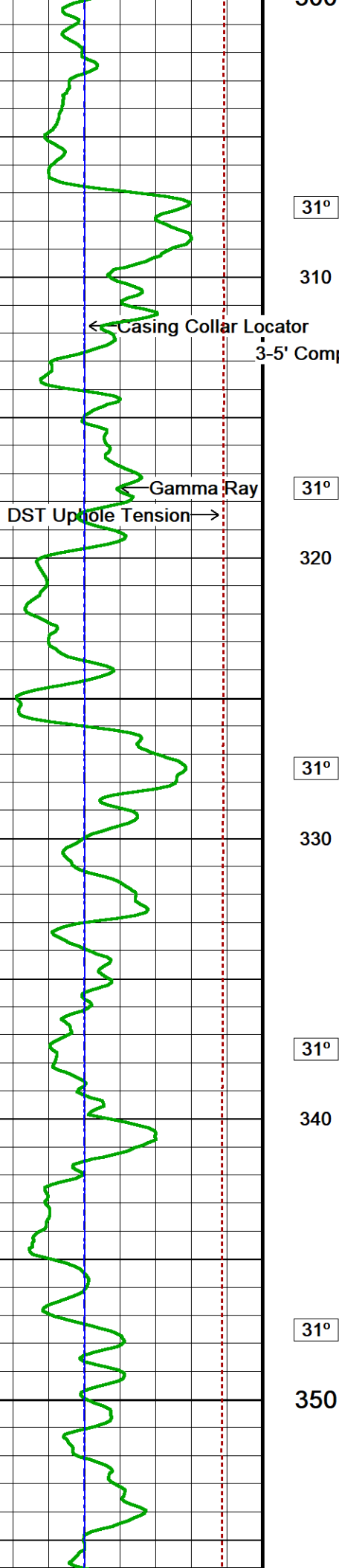
31°

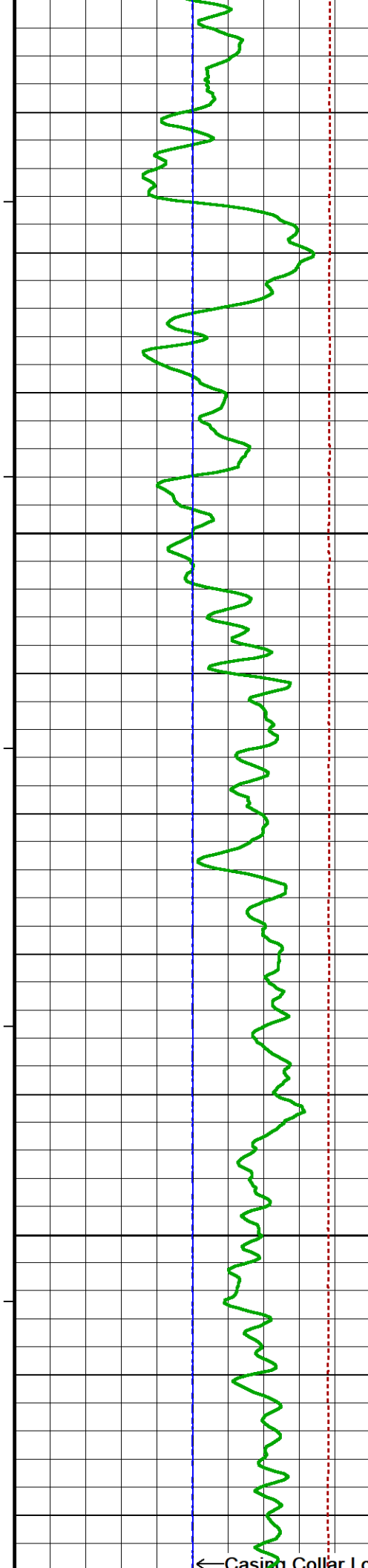
290

31°

300







31°

360

31°

370

31°

380

31°

390

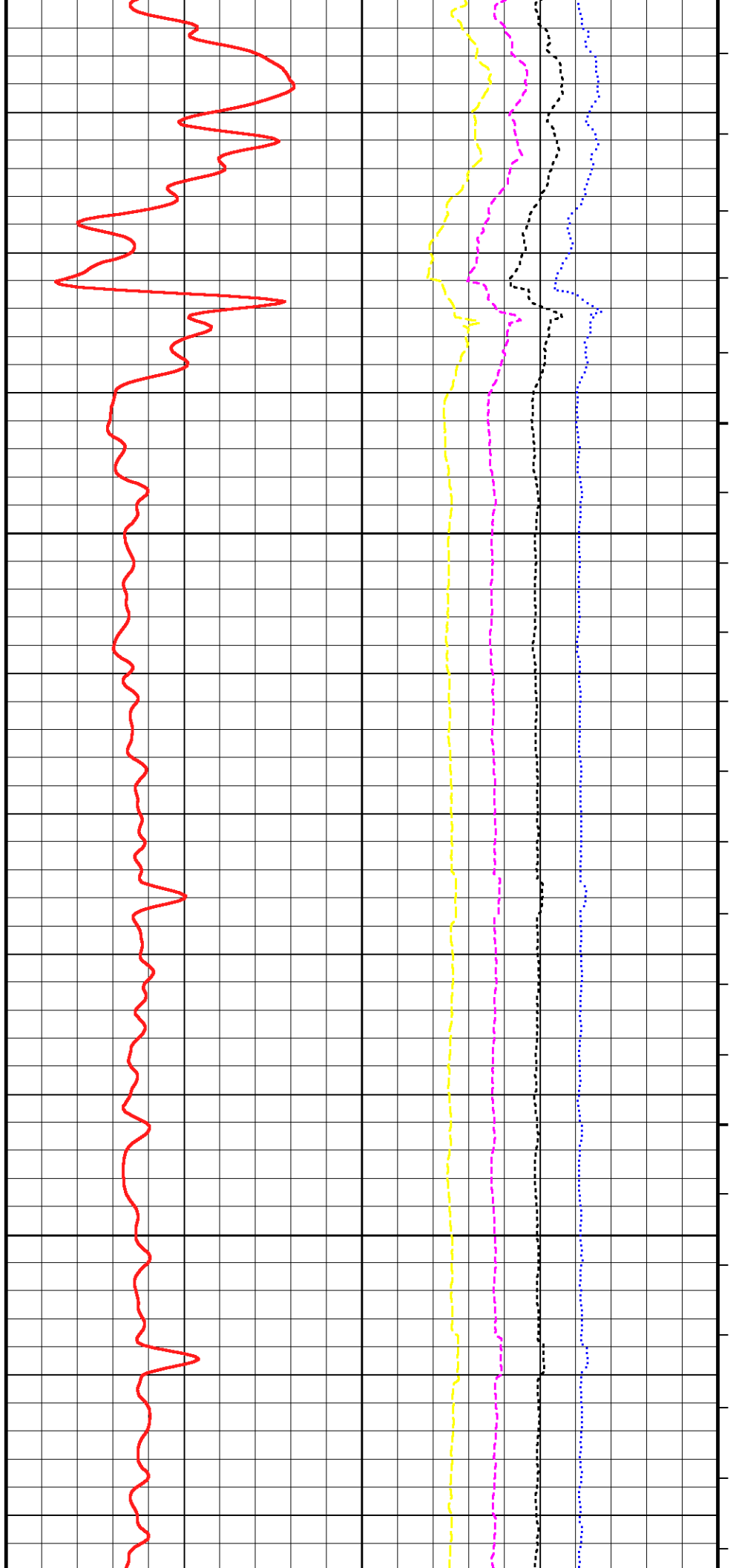
31°

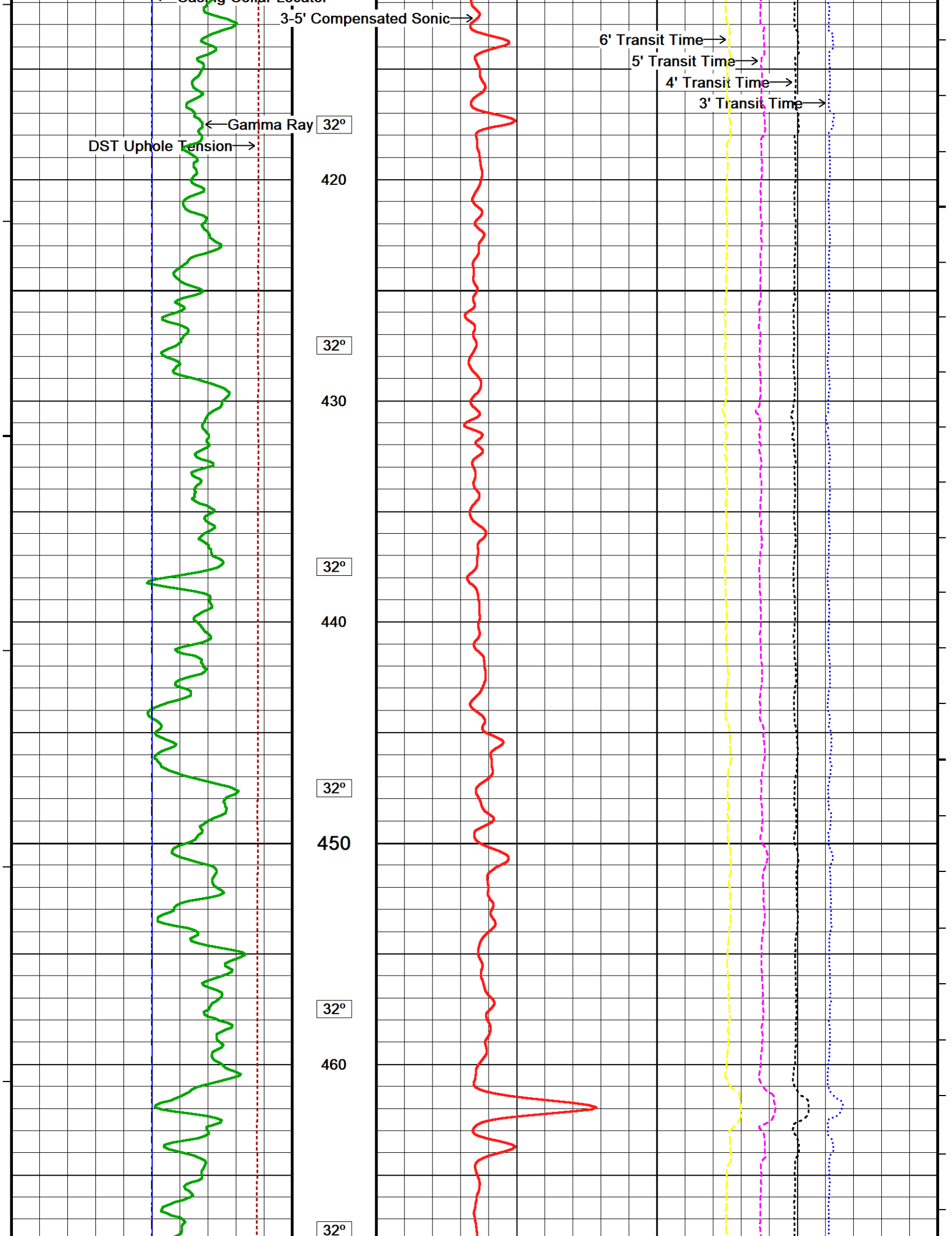
400

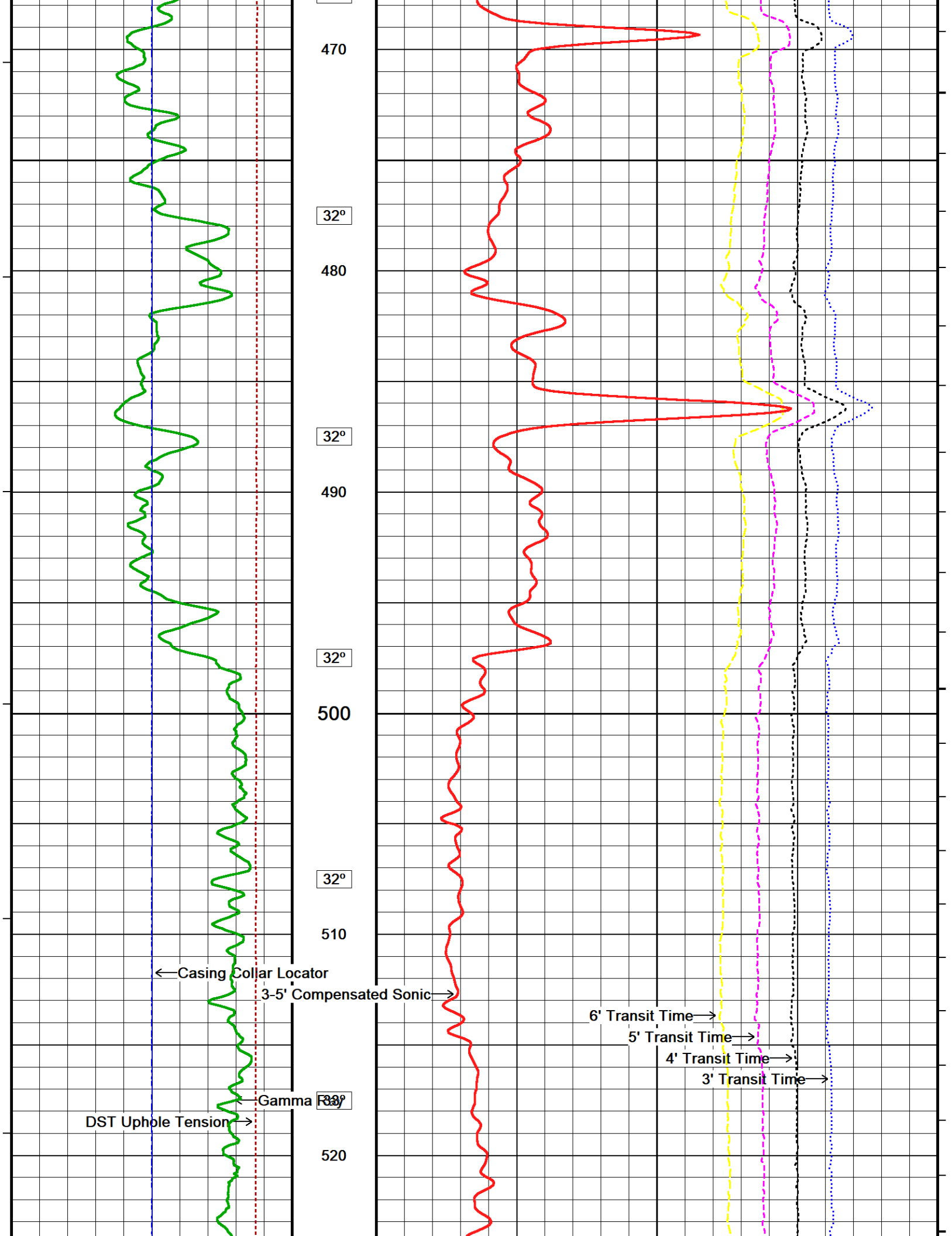
31°

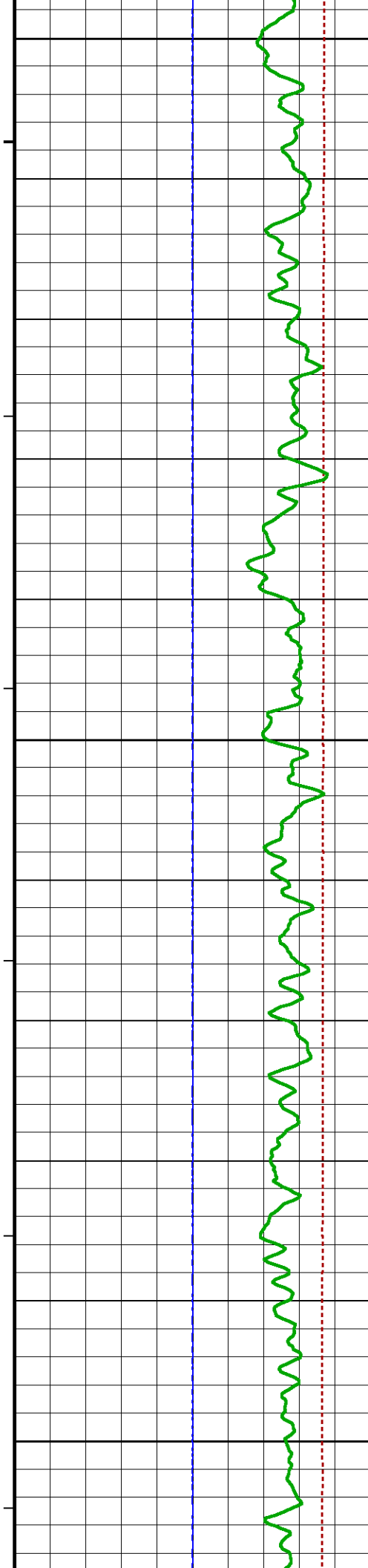
410

← Casino Collar Locator









33°

530

33°

540

33°

550

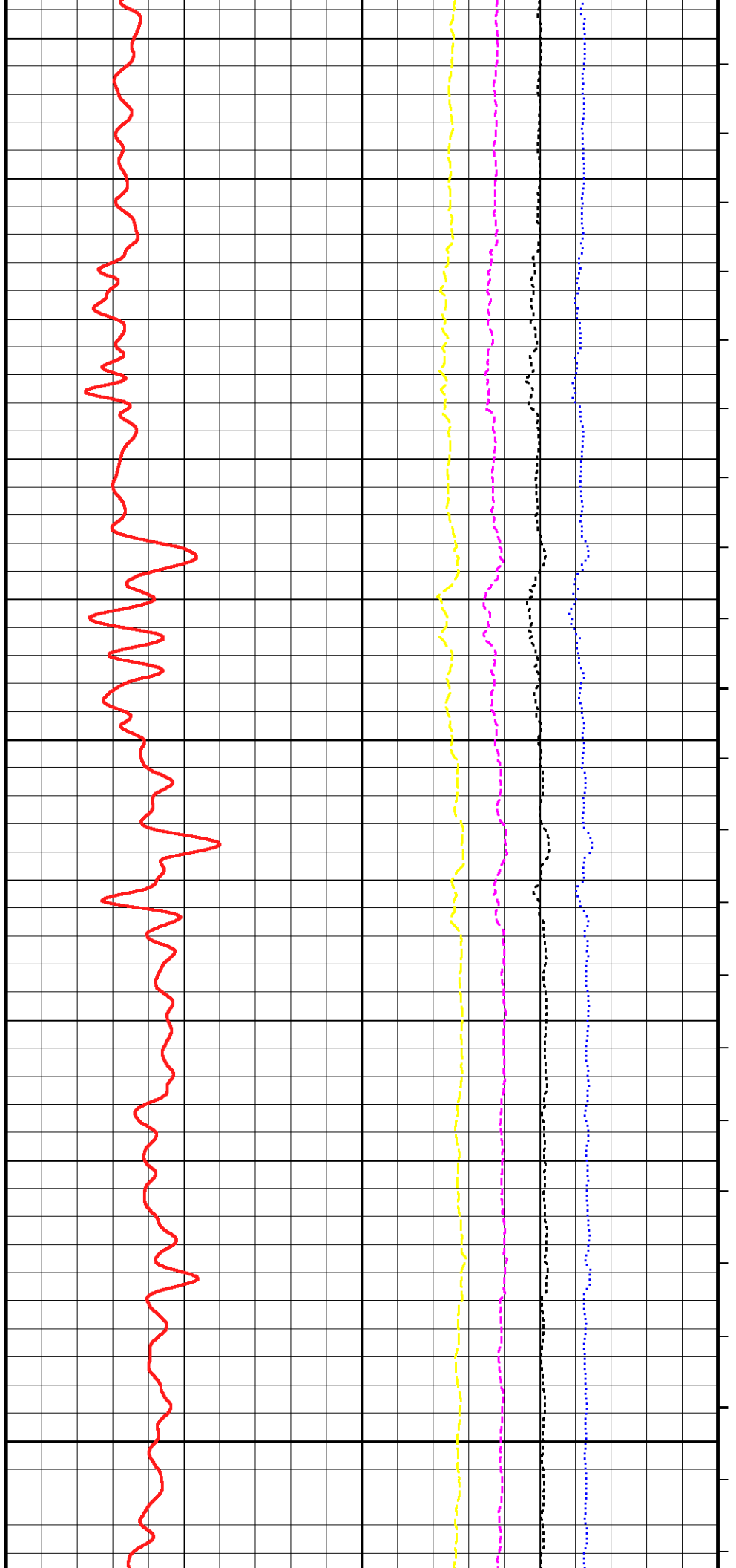
33°

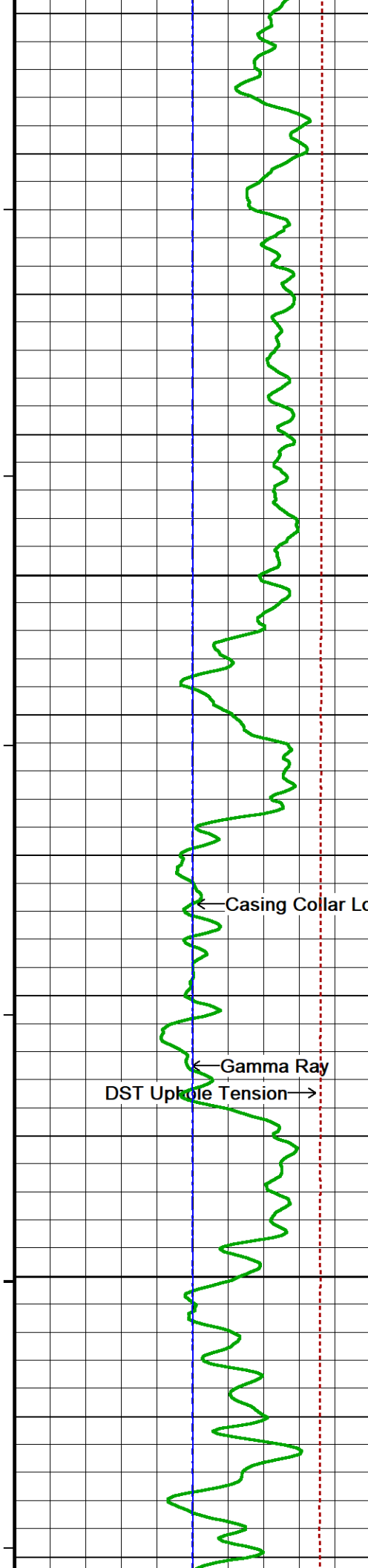
560

33°

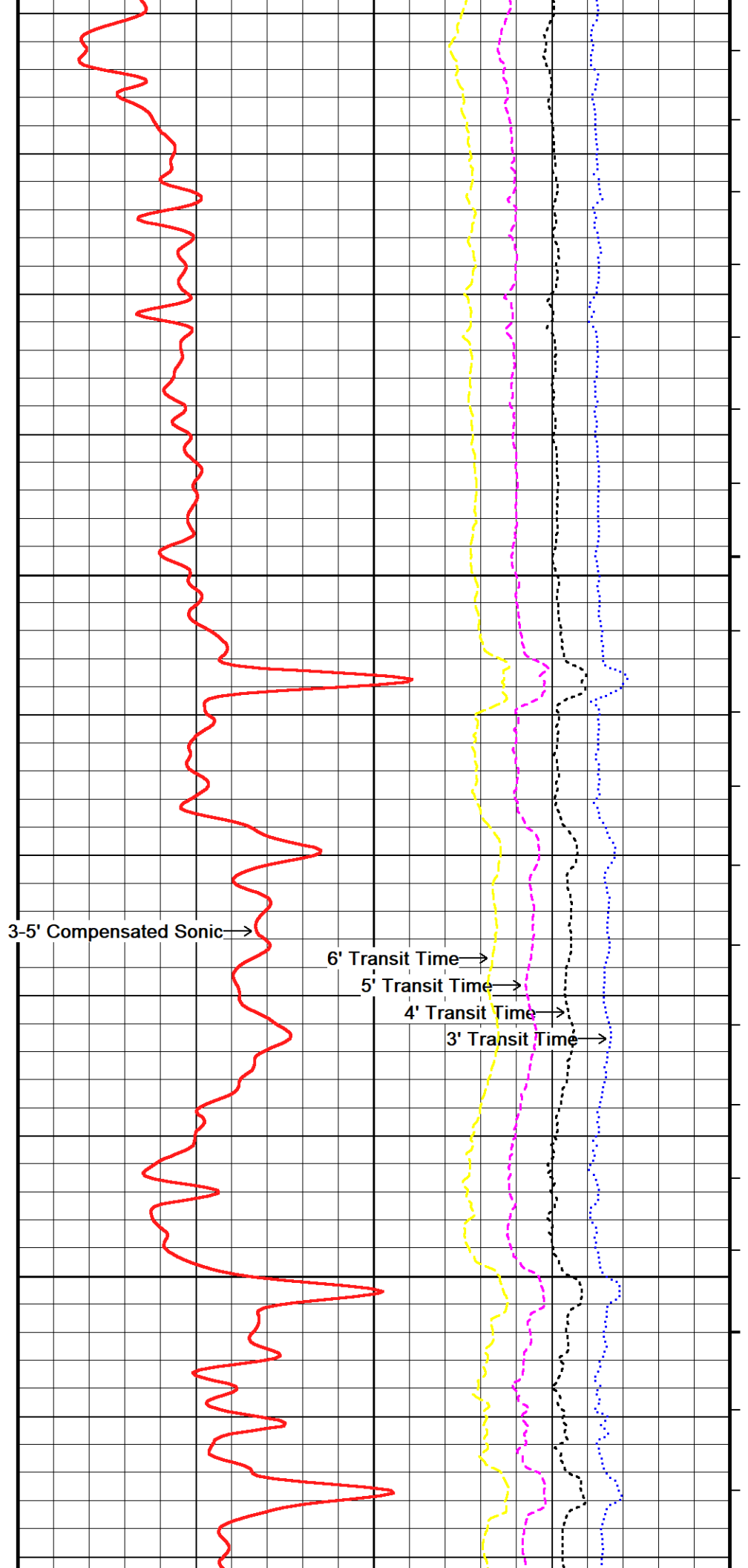
570

33°



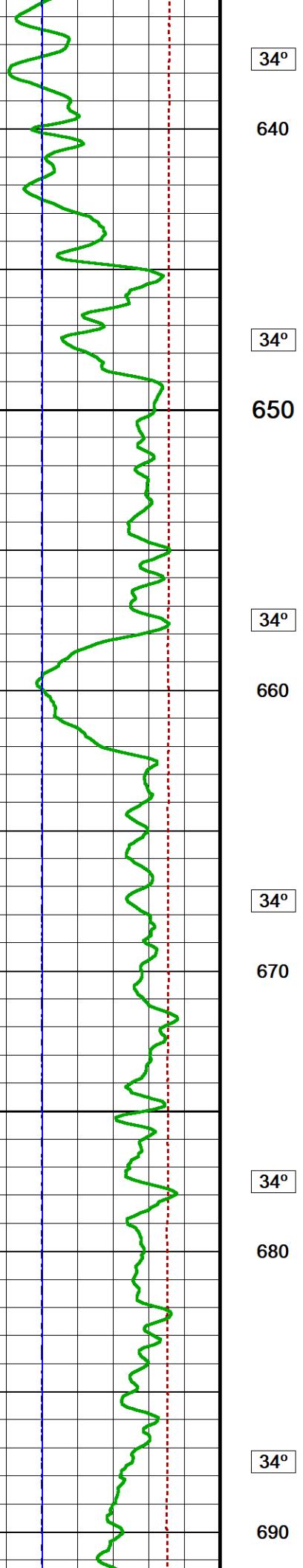


580
33°
590
33°
600
33°
610
33°
620
34°
630

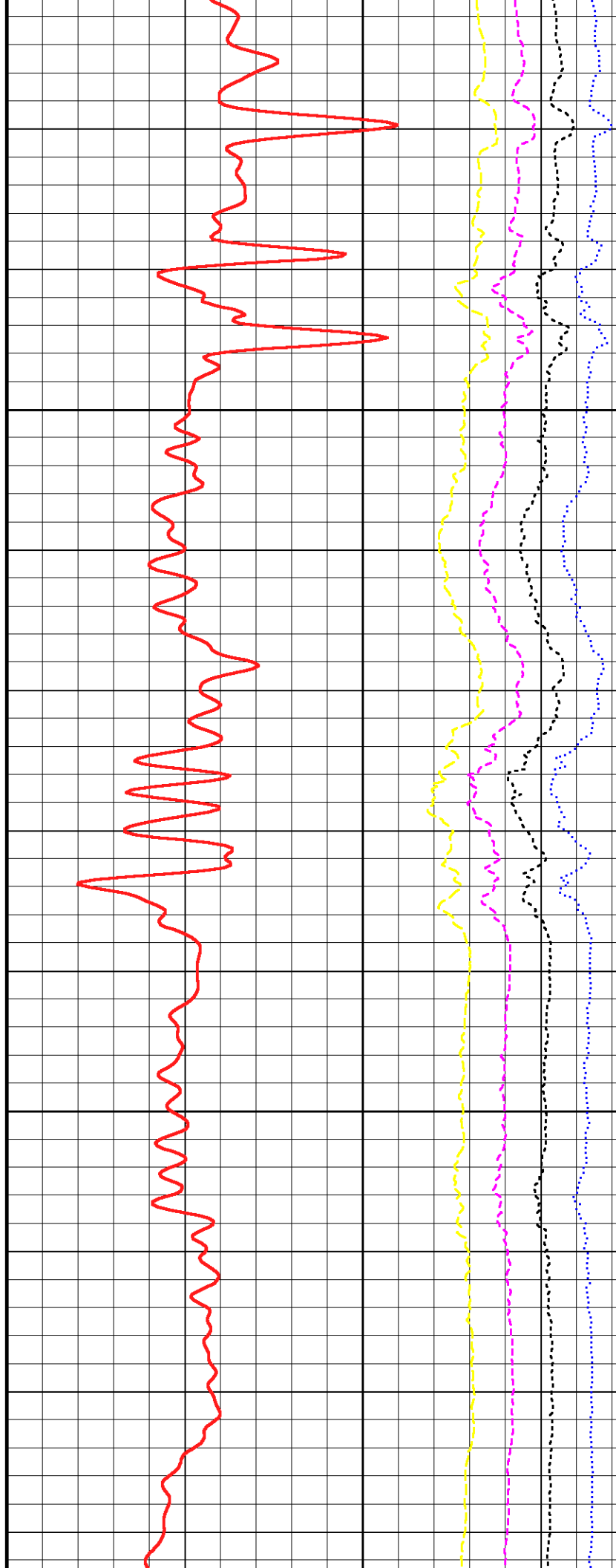


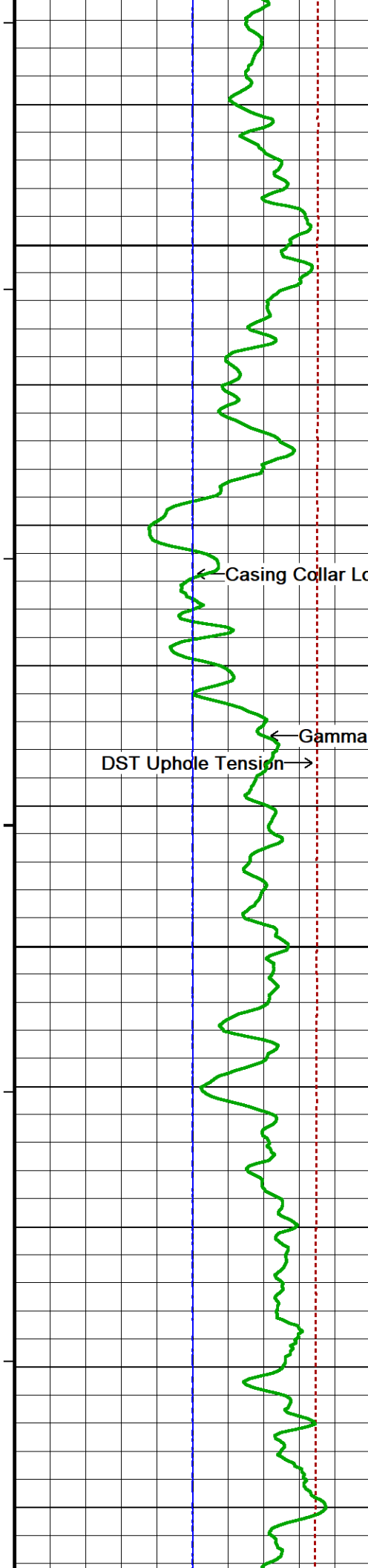
← Casing Collar Locator
← Gamma Ray
DST Up-hole Tension →

3-5' Compensated Sonic →
6' Transit Time →
5' Transit Time →
4' Transit Time →
3' Transit Time →



34° 640 34° 650 34° 660 34° 670 34° 680 34° 690





34°

700

35°

710

Casing Collar Locator

35°

Gamma Ray

DST Uphole Tension

720

35°

730

35°

740

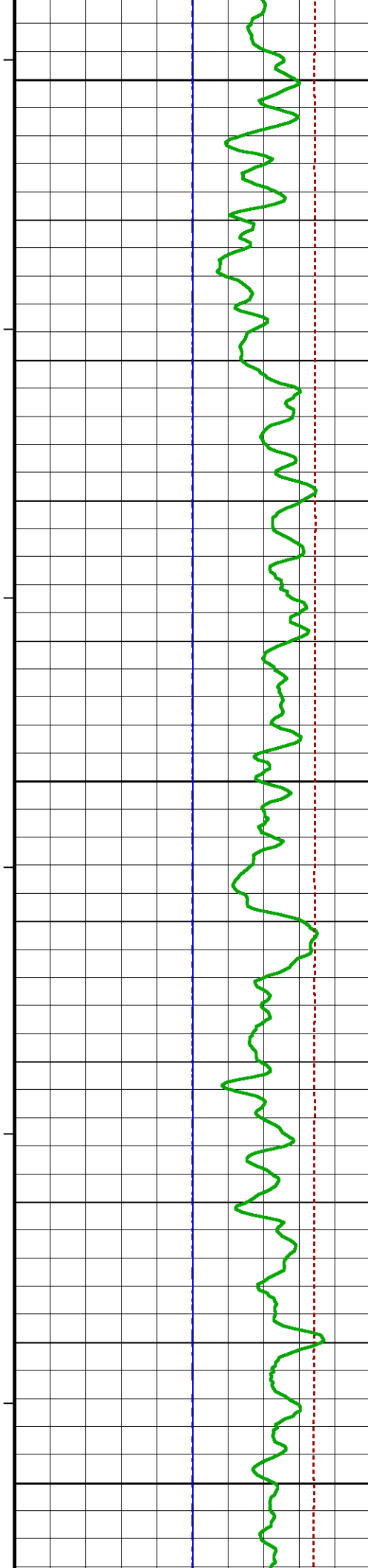
3-5' Compensated Sonic

6' Transit Time

5' Transit Time

4' Transit Time

3' Transit Time



35°

750

35°

760

35°

770

35°

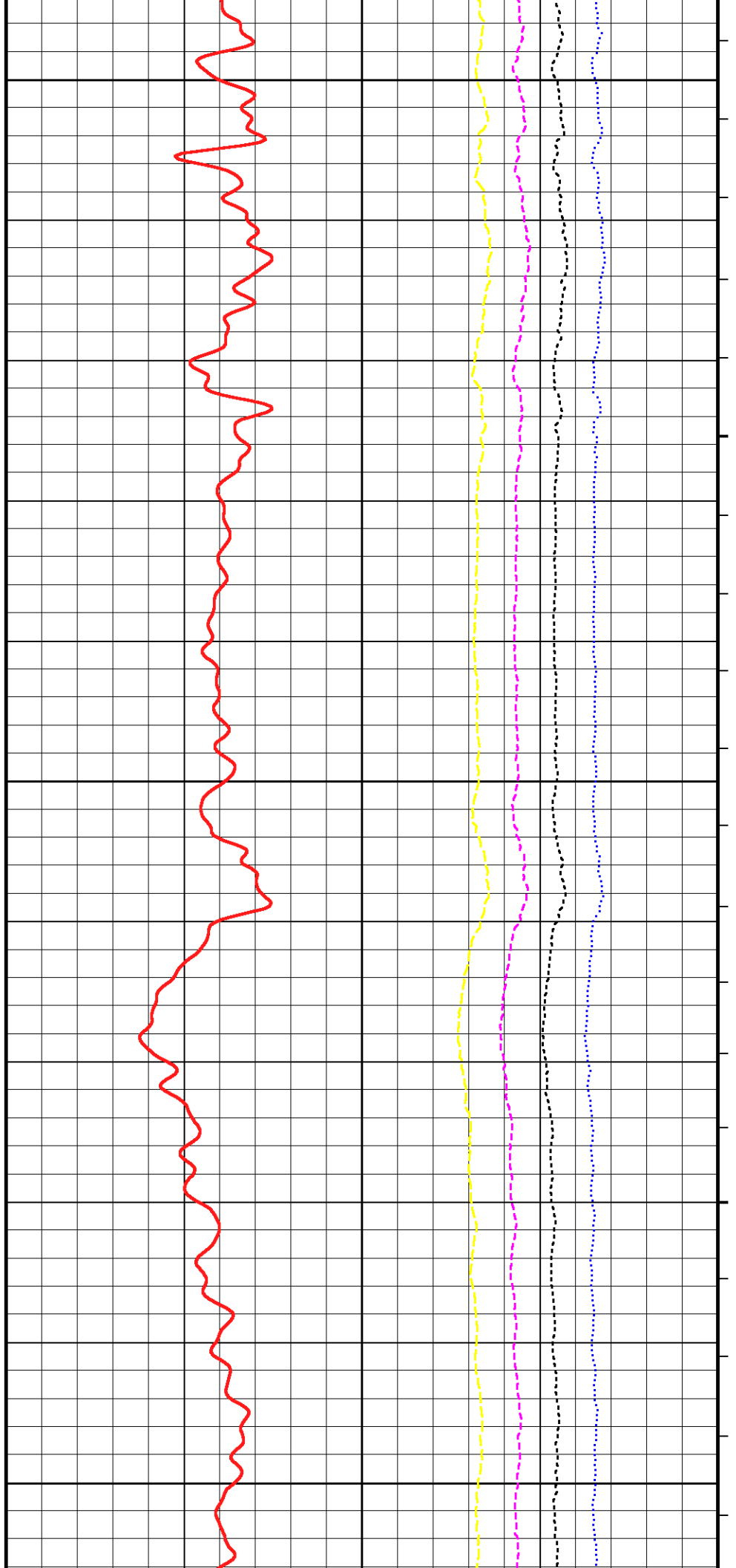
780

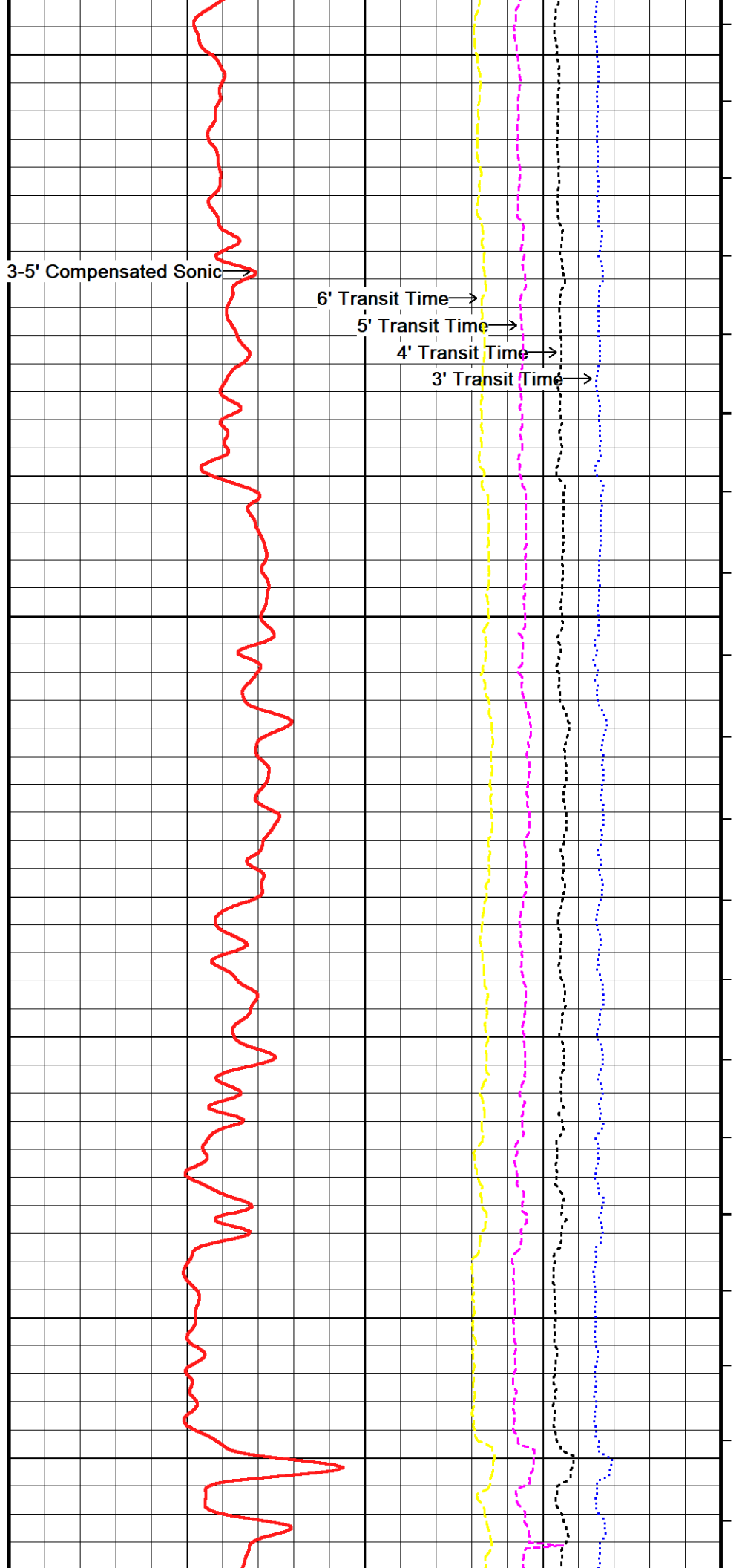
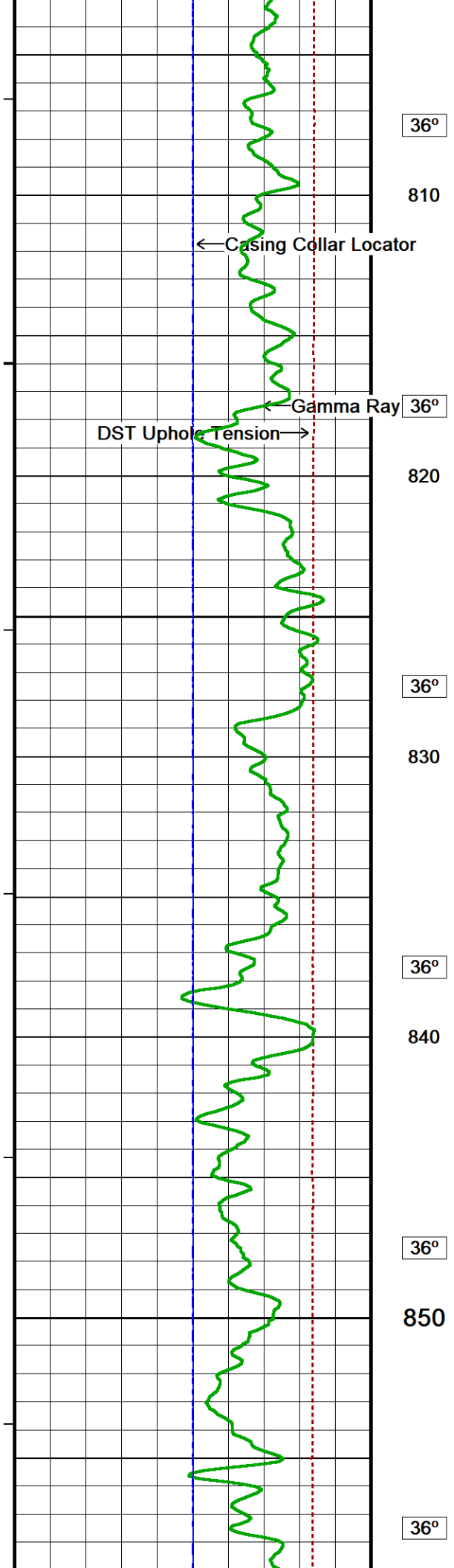
35°

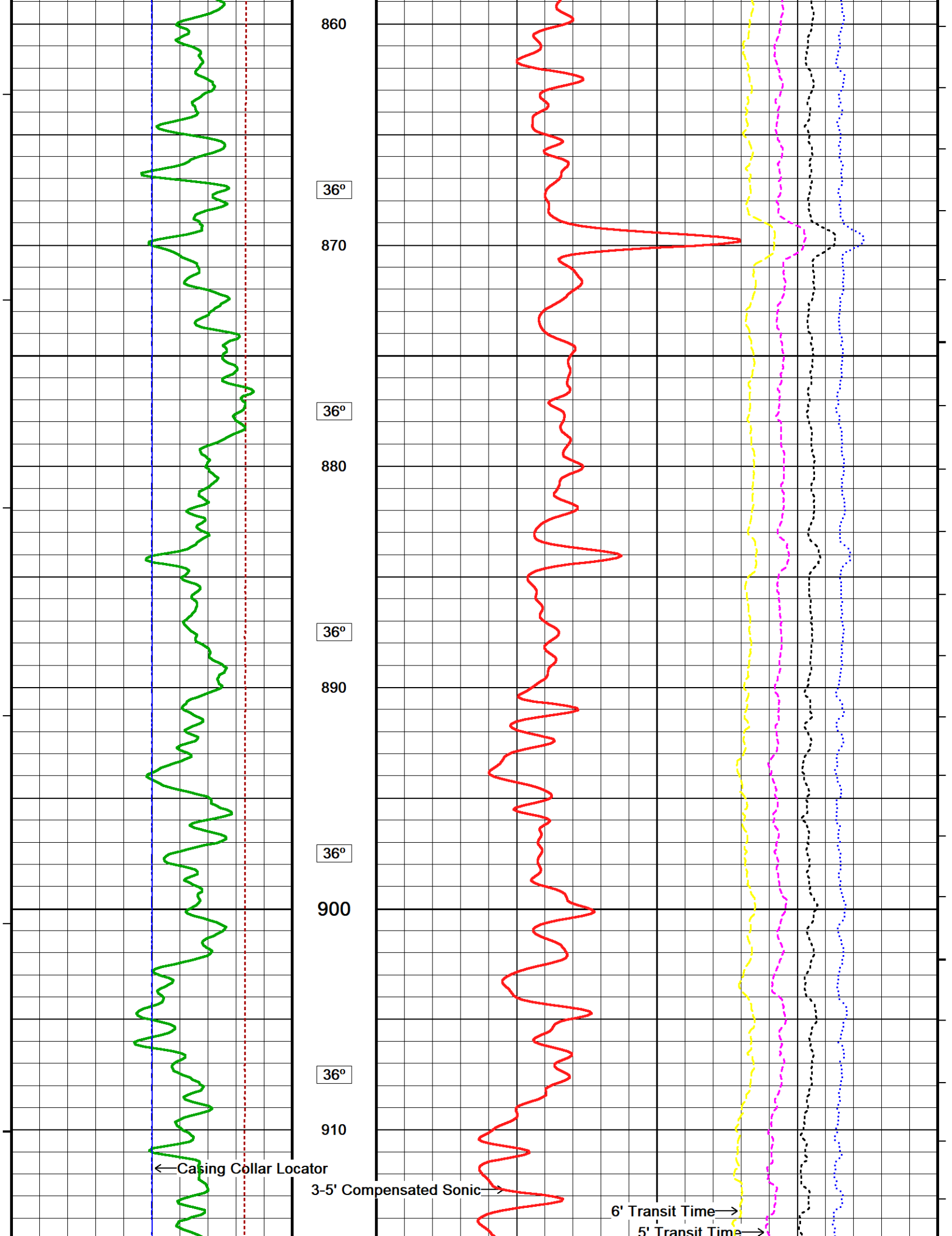
790

36°

800







DST Uphole Tension →

← Gamma Ray 37°

920

37°

930

37°

940

37°

950

37°

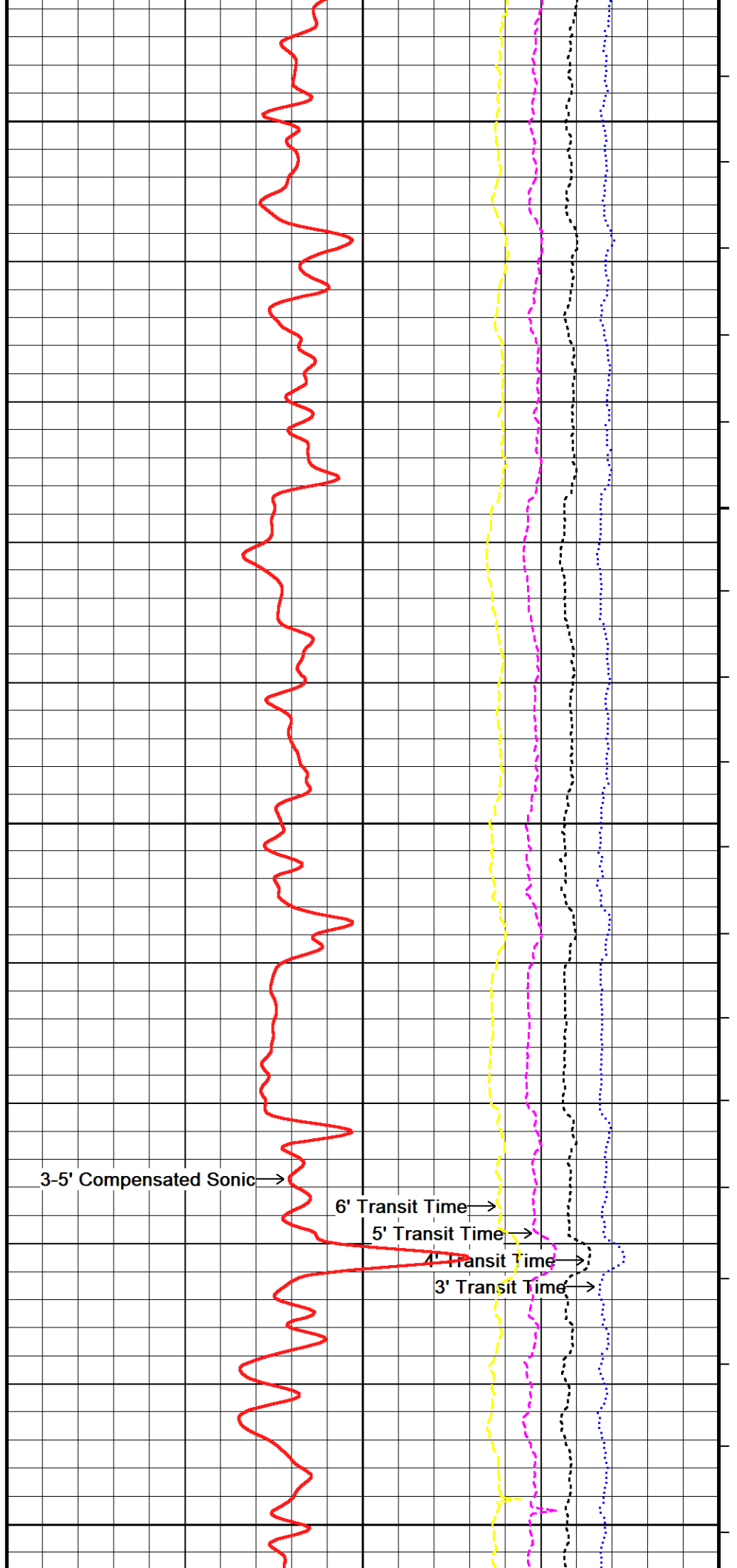
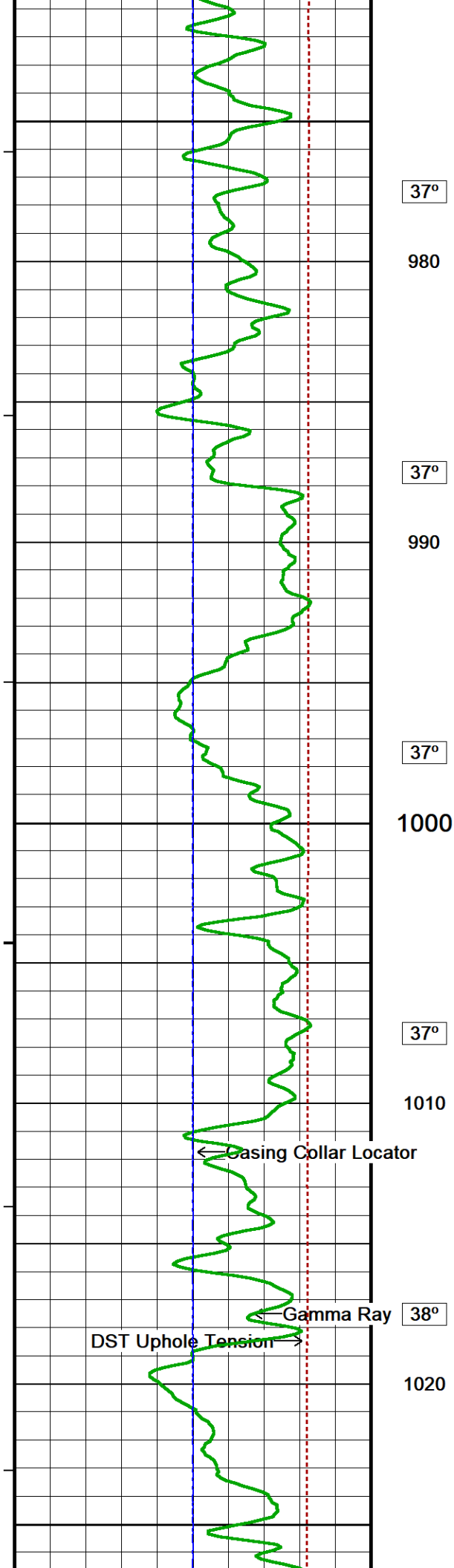
960

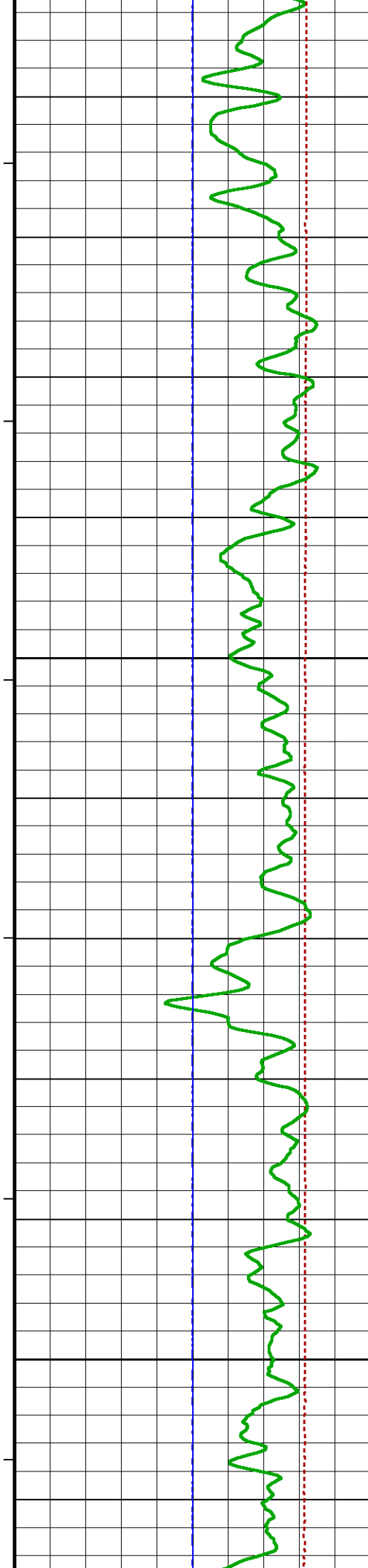
37°

970

4' Transit Time →

3' Transit Time →





38°

1030

38°

1040

38°

1050

38°

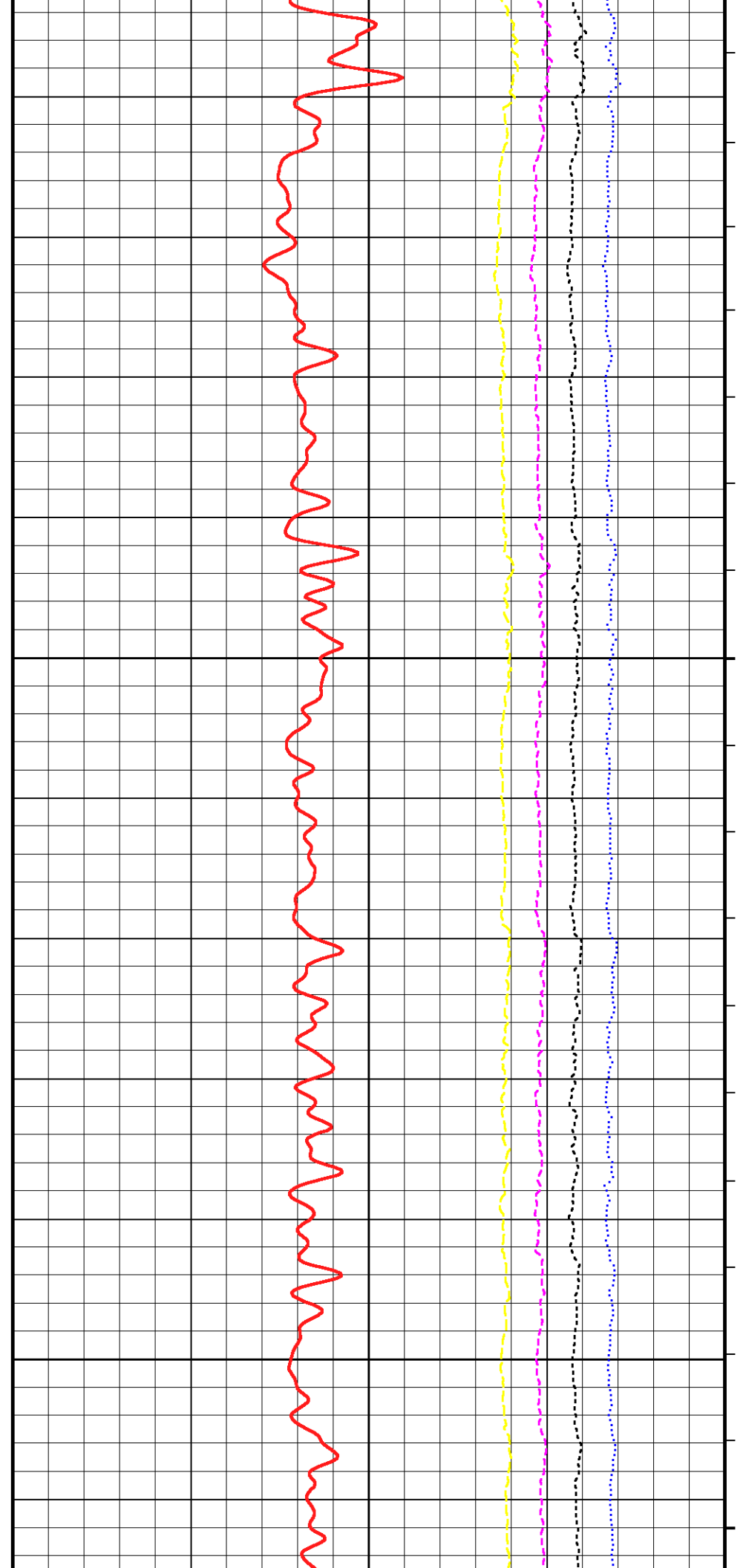
1060

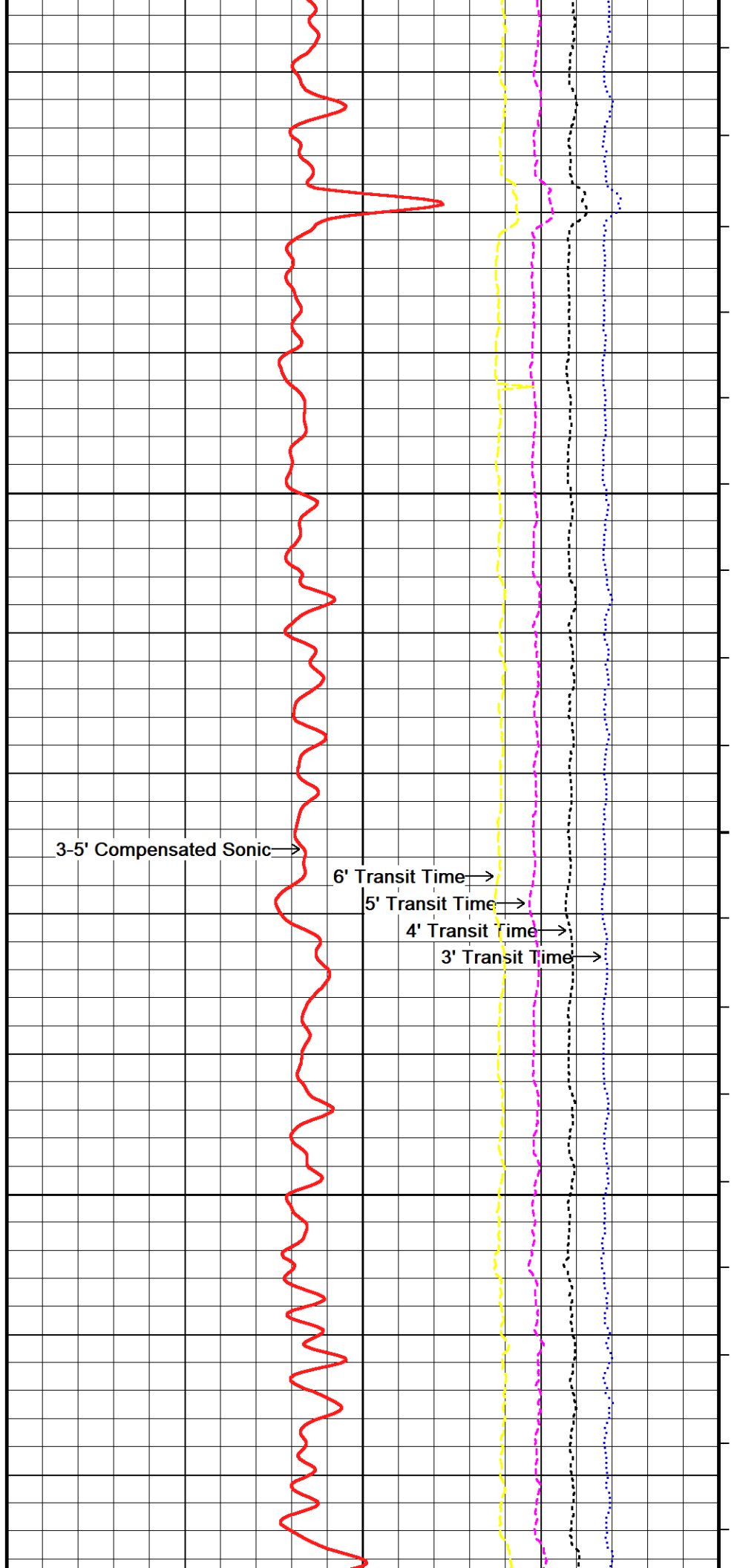
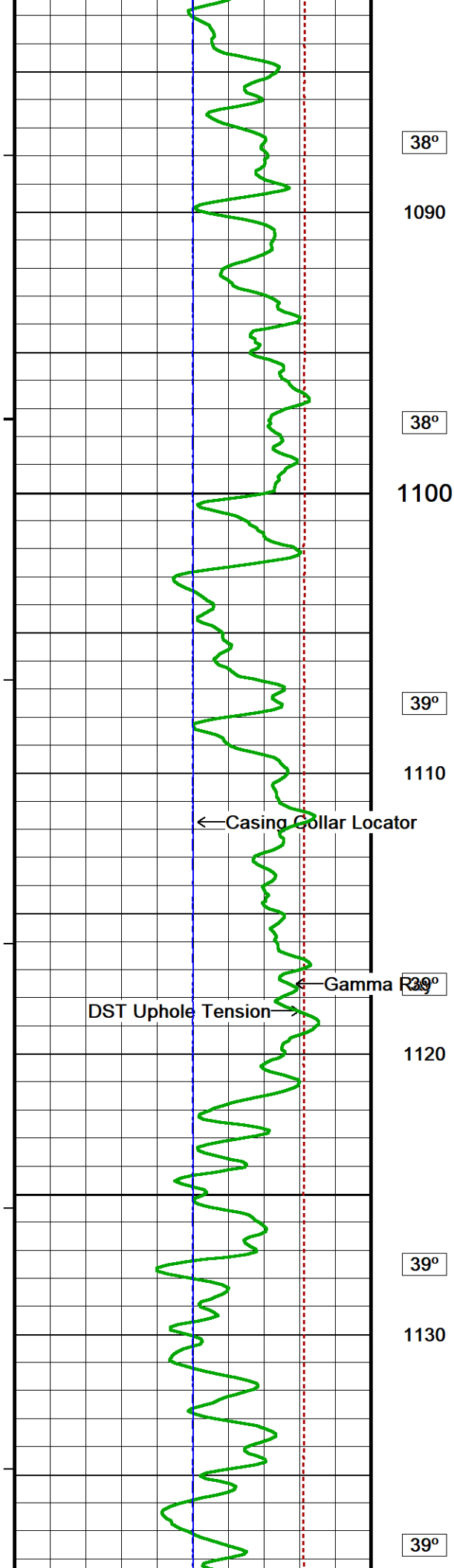
38°

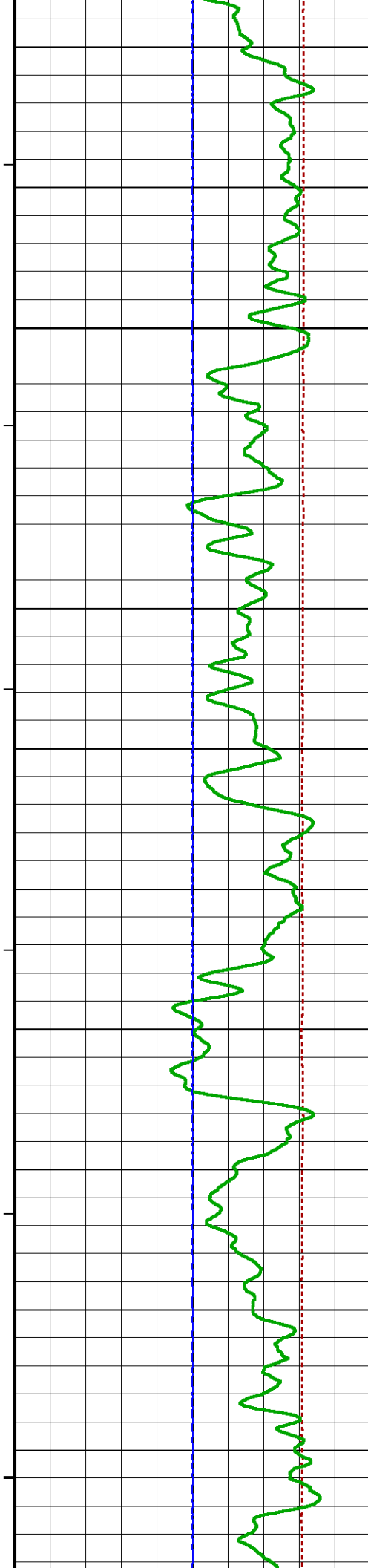
1070

38°

1080







1140

39°

1150

39°

1160

39°

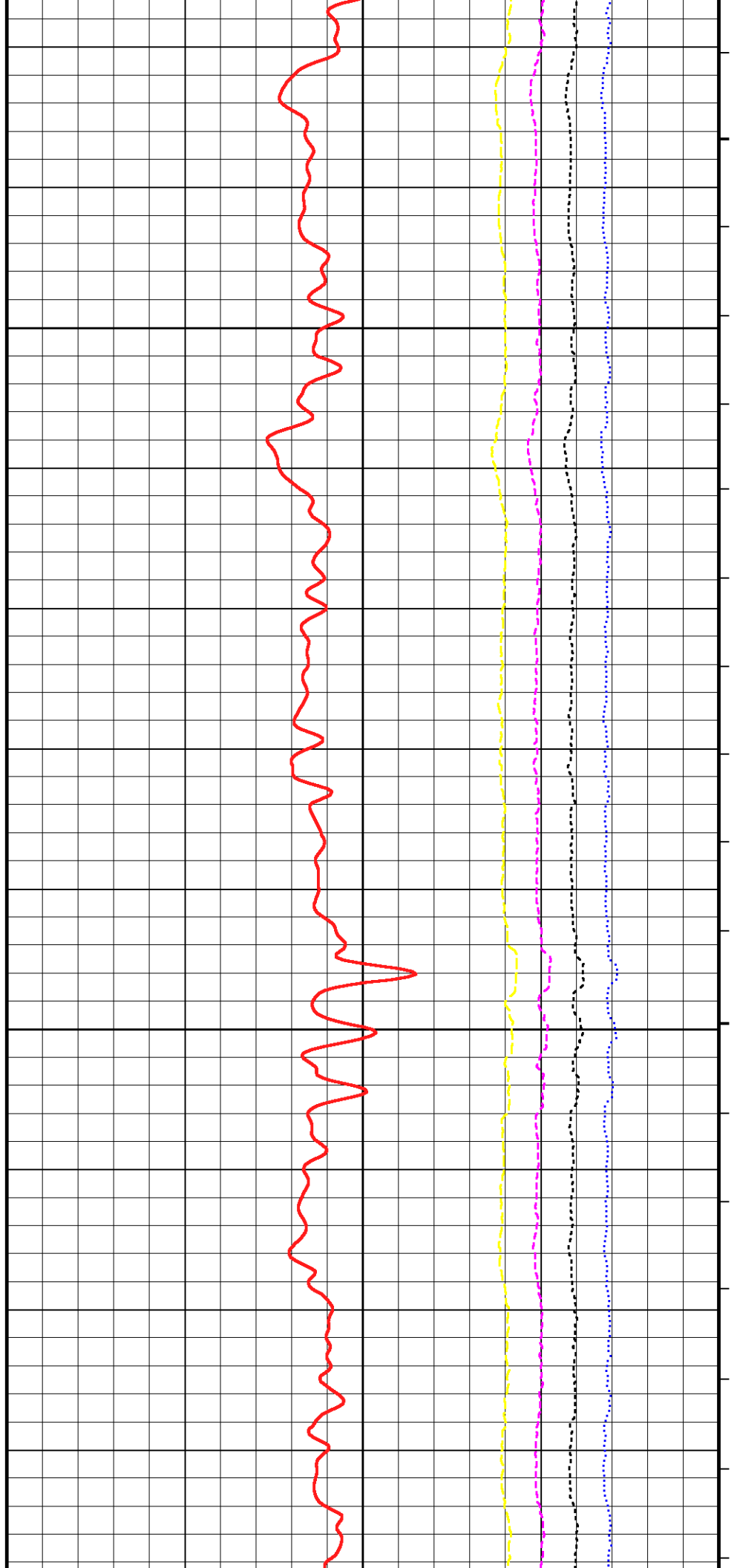
1170

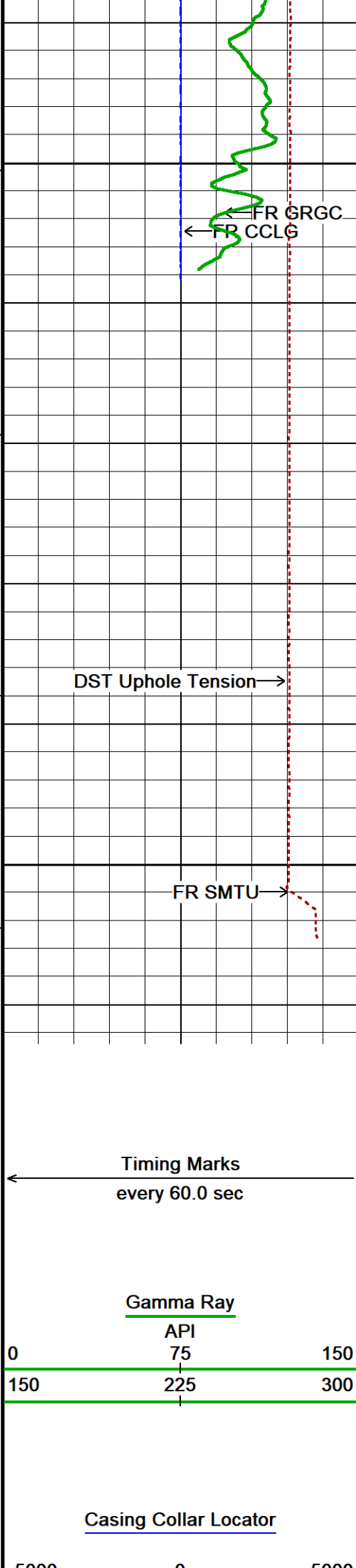
39°

1180

39°

1190





39°

1200

1210

1220

TD

1230

Depth in Metres

3-5' Compensated Sonic
microsec/foot

140

115

90

65

40

3' Transit Time
microseconds

1100

600

100

4' Transit Time
microseconds

1100

600

100

5' Transit Time
microseconds

1100

600

100

← FR GRGC
← FR CCLG

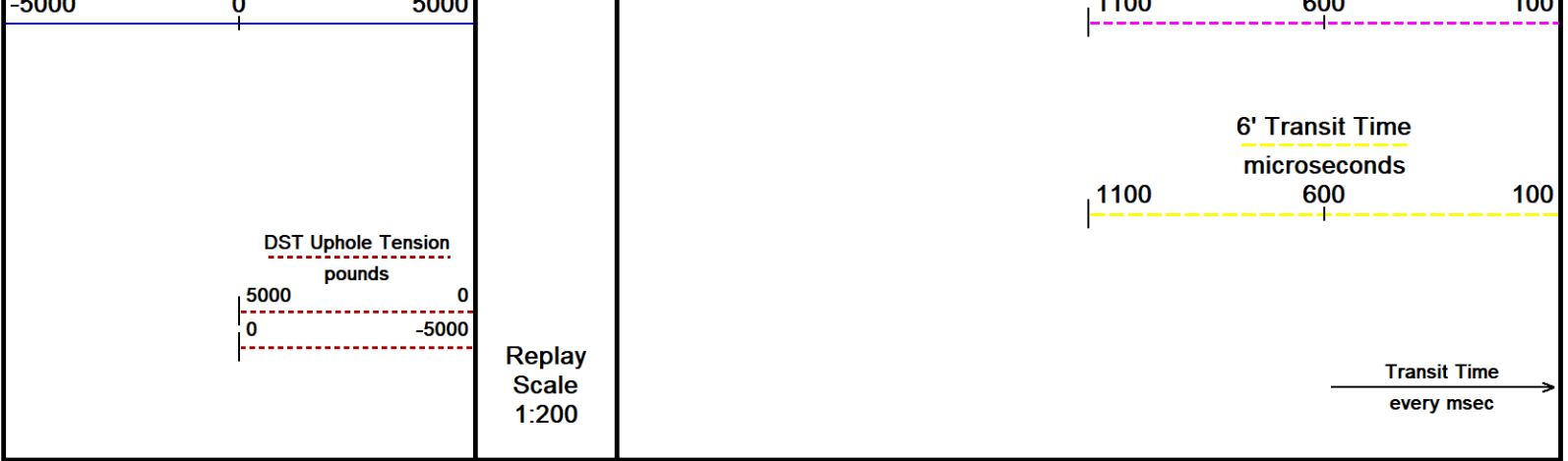
DST Uphole Tension→

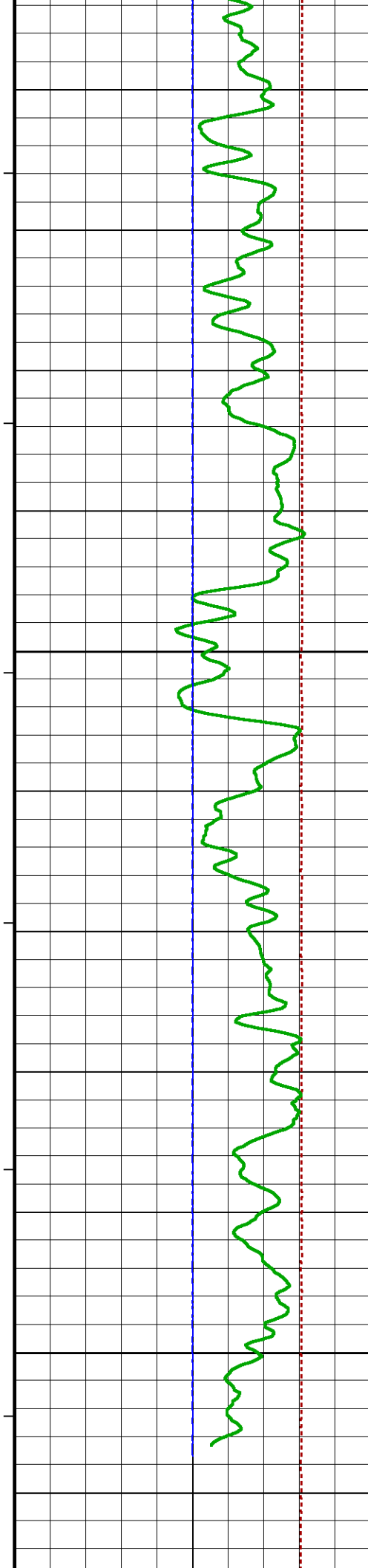
FR SMTU→

FR DT35→

FR FR TR22→

FR FR TR21→





39°

1160

39°

1170

39°

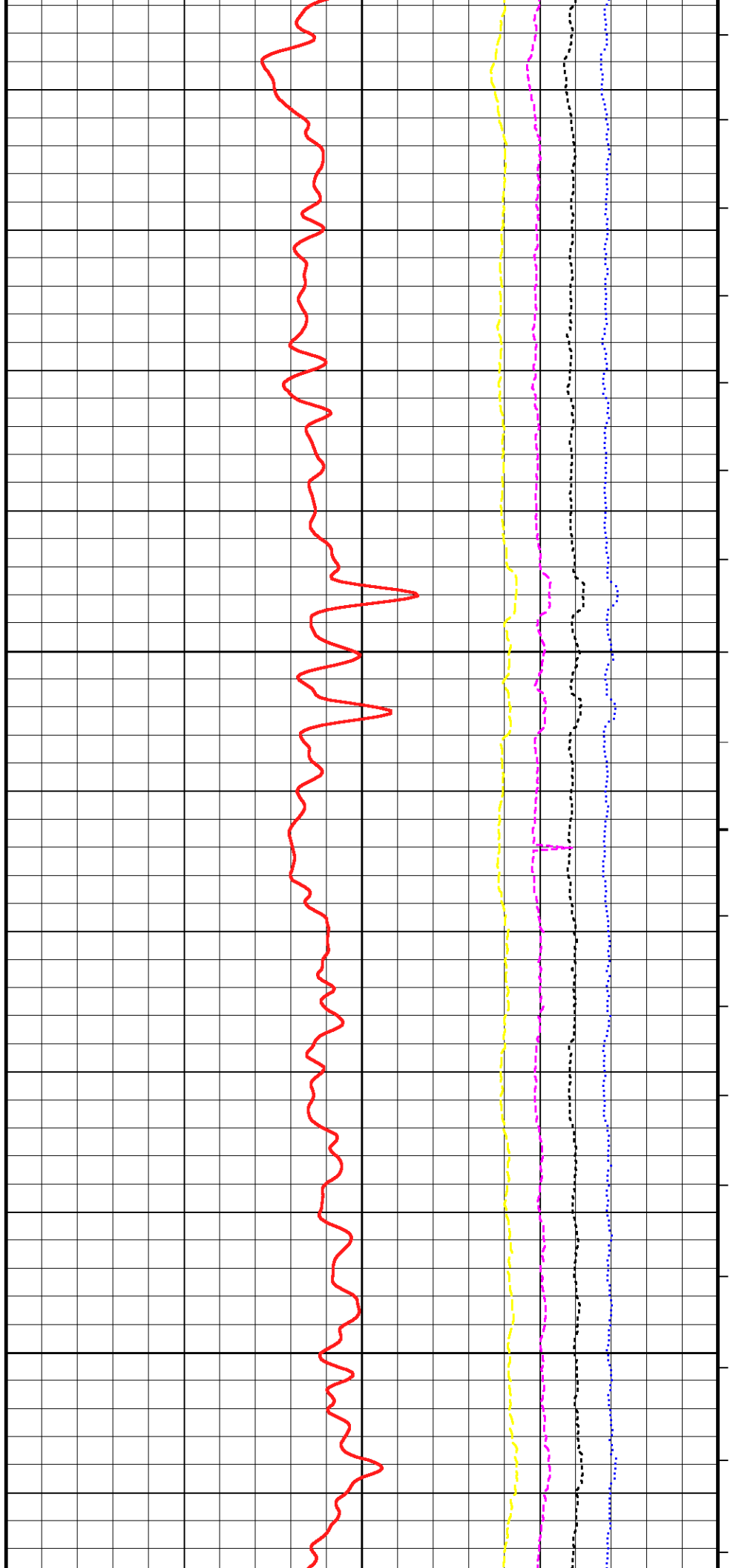
1180

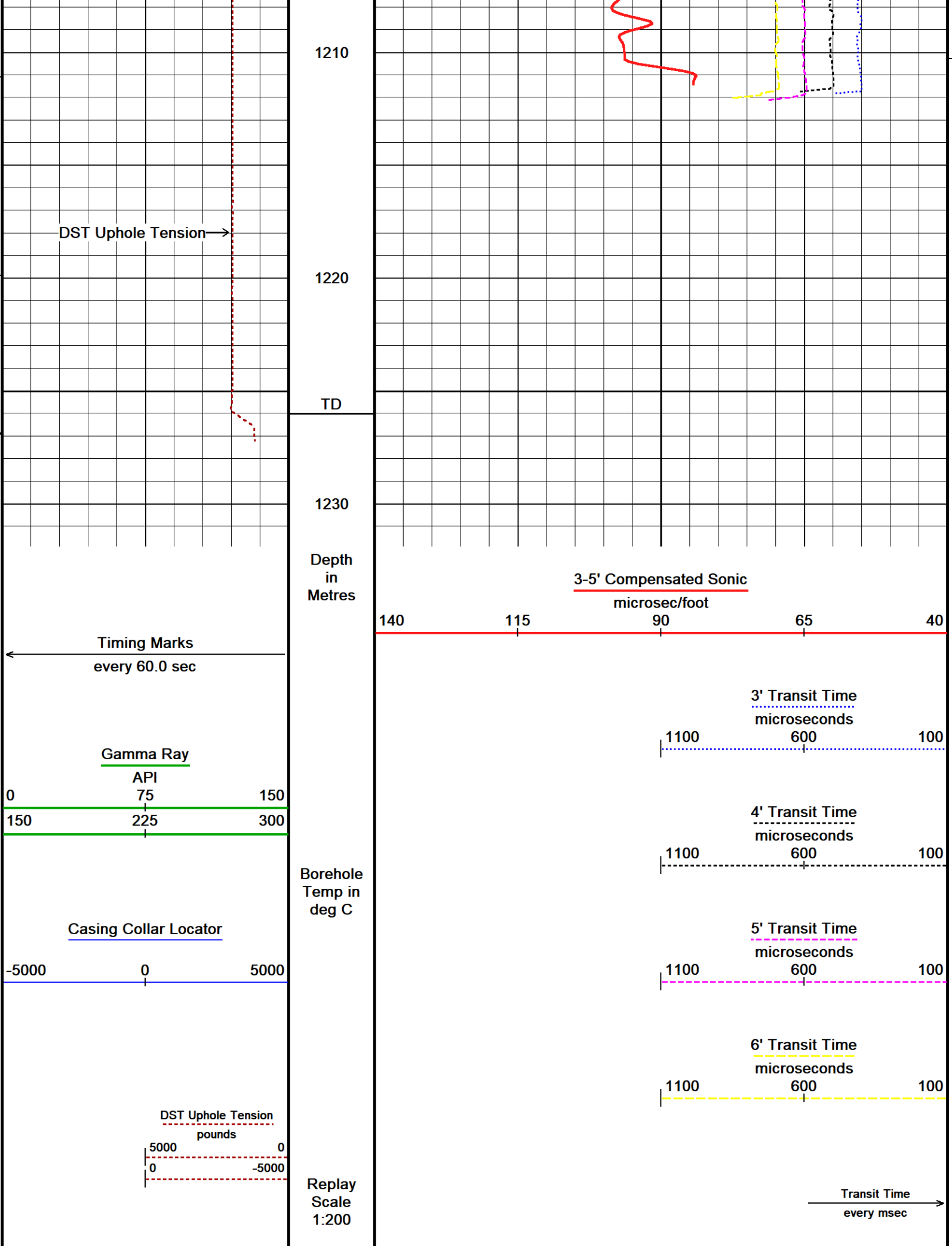
38°

1190

38°

1200





BEFORE SURVEY CALIBRATION
C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView8d\0\209_NADES_RES_CALB.dta

General Constants All 000Last Edited on 04-SEP-2024,06:18

General Parameters

Mud Resistivity0.150ohm-metres

Mud Resistivity Temperature25.000degrees C

Water Level0.000metres

Borehole Fluid ProcessingWater Level Switch

Hole/Annular Volume and Differential Caliper Parameters

HVOL MethodXY Caliper

HVOL Caliper 1X Two Arm Caliper

HVOL Caliper 2Y Two Arm Caliper

Annular Volume Diameter5.500inches

Caliper for Differential CaliperNone

Rwa Parameters

Porosity usedWyllie Lmst Sonic Por.

Resistivity usedDeep Laterolog

RWA Constant A0.620

RWA Constant M2.150

SW/APOR Tool Source0.000

Gamma Calibration MCG-E.A 531Field Calibration on 04-SEP-2024 05:54

MeasuredCalibrated (API)

Background7048

Calibrator (Gross)1092755

Calibrator (Net)1022707

Gamma Calibration Tolerances MCG-E.A 531

Ratio1.446

1.401.4751.55

Counts/API

Gamma Constants MCG-E.A 531Last Edited on 04-SEP-2024,06:18

Gamma Calibrator Number159

GRC-M Calibrator Jig in Use?NO

Inactive Background Jig in Use?NO

Mud Density1.10gm/cc

Caliper Source for ProcessingY Two Arm Caliper

Tool PositionCentred

Potassium EquivalenceChloride

K Mud Concentration0.00%

High Resolution Temperature Calibration MCG-E.A 531Field Calibration on 03-SEP-2024,09:41

MeasuredCalibrated(Deg C)

Lower0.000.00

Upper100.00100.00

High Resolution Temperature Constants MCG-E.A 531Last Edited on 03-SEP-2024,09:41

Pre-filter Length11

Accelerometer Parameters MBN-C.A 55

Date Of Last Accelerometer Calibration20-NOV-2023,13:27

X AccelerometerY AccelerometerZ Accelerometer

Slope-1.084180-1.103980-1.114574

Offset0.0027100.0063610.008729

Accelerometer Constants MBN-C.A 55

Accelerometer Calibrator Number		000			
Accelerometer Temperature Characterisation					
X Accelerometer					
Serial Number	0				
Calibration Date	01-Jan-1998				
	B0	B1	B2	B3	
Bias(g)	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
	SF0	SF1	SF2	SF3	
Scale Factor(mA/g)	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
Y Accelerometer					
Serial Number	0				
Calibration Date	01-Jan-1998				
	B0	B1	B2	B3	
Bias(g)	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
	SF0	SF1	SF2	SF3	
Scale Factor(mA/g)	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
Z Accelerometer					
Serial Number	0				
Calibration Date	01-Jan-1998				
	B0	B1	B2	B3	
Bias(g)	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
	SF0	SF1	SF2	SF3	
Scale Factor(mA/g)	0.00000e+00	0.00000e+00	0.00000e+00	0.00000e+00	
Navigation Constants MBN-C.A 55				Last Edited on 04-SEP-2024,06:19	
Magnetic Declination		6.53	degrees	East	
Magnetometer Parameters MBN-C.A 55					
Date Of Last Magnetometer Calibration		22-MAY-2024,11:04			
	X Magnetometer	Y Magnetometer	Z Magnetometer		
Slope	-1.000000	1.002324	1.004838		
Offset	0.006984	0.017764	-0.012985		
Magnetometer Constants MBN-C.A 55					
Magnetometer Calibrator Number		000			
Sonic Constants MSS-D.A 446				Last Edited on 03-SEP-2024,09:42	
Maximum Boundary Contrast		70.00	micro-sec/ft		
Fluid Transit Time		189.00	micro-sec/ft		
Limestone Transit Time		47.50	micro-sec/ft		
Sandstone Transit Time		55.50	micro-sec/ft		
Dolomite Transit Time		43.50	micro-sec/ft		
Sonic used for Porosities		3-5' Compensated Sonic			
Correction for Sonde Skew		Applied			
Cycle Stretch Algorithm		Applied			
MN3FT		N/A	micro-sec		
MX3FT		N/A	micro-sec		
Hunt-Raymer Constant		83.12	micro-sec/ft		
Sonde Mode		Compensated			
Hole Type		Open Hole			
Sonde Parameters					
	Measured	Calibrated			
Offset		0.0000			
Free Pipe	N/A	N/A			
Peak Amplitude Source		N/A			
Waveform	Start Time (micro-sec)	Width (micro-sec)	Pre Gain	Start Gain	Discriminator (mV)
3'	N/A	N/A	N/A	N/A	N/A
4'	N/A	N/A	N/A	N/A	N/A
5'	N/A	N/A	N/A	N/A	N/A
6'	N/A	N/A	N/A	N/A	N/A
Processed Fixed Gate Parameters					

Waveform Used For Processing	N/A		
Start Time (micro-sec)	End Time (micro-sec)	Discriminator (mV)	N/A
N/A	N/A	N/A	
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A
N/A	N/A	N/A	N/A

Full Waveform Parameters

Use 3' Waveform to derive TR	N/A	
Use 4' Waveform to derive TR	N/A	
Use 5' Waveform to derive TR	N/A	
Use 6' Waveform to derive TR	N/A	
3' Waveform Discriminator Level	N/A	mV
4' Waveform Discriminator Level	N/A	mV
5' Waveform Discriminator Level	N/A	mV
6' Waveform Discriminator Level	N/A	mV
Waveform Discriminator Filter	N/A	
Semblance Window Width	N/A	micro-sec
Semblance Processing Enabled	N/A	
Tracking Boxes Enabled In Processing	N/A	

DOWNHOLE EQUIPMENT

C:\Users\sysadmin\AppData\Local\Temp\Weatherford PreView8d\0\209_NADES_RES_CALB.dta

Cablehead, 11 pin
CBH-CC 401 LG: 0.73 m WT: 24.3 lb OD: 2.240 in

11C-11B Compact Tool Adaptor
MTA-K.B 309 LG: 0.47 m WT: 13.2 lb OD: 2.240 in

Compact Stiff Bridle Electrode Sub.
MBE-C.A 154 LG: 3.76 m WT: 77.2 lb OD: 2.244 in

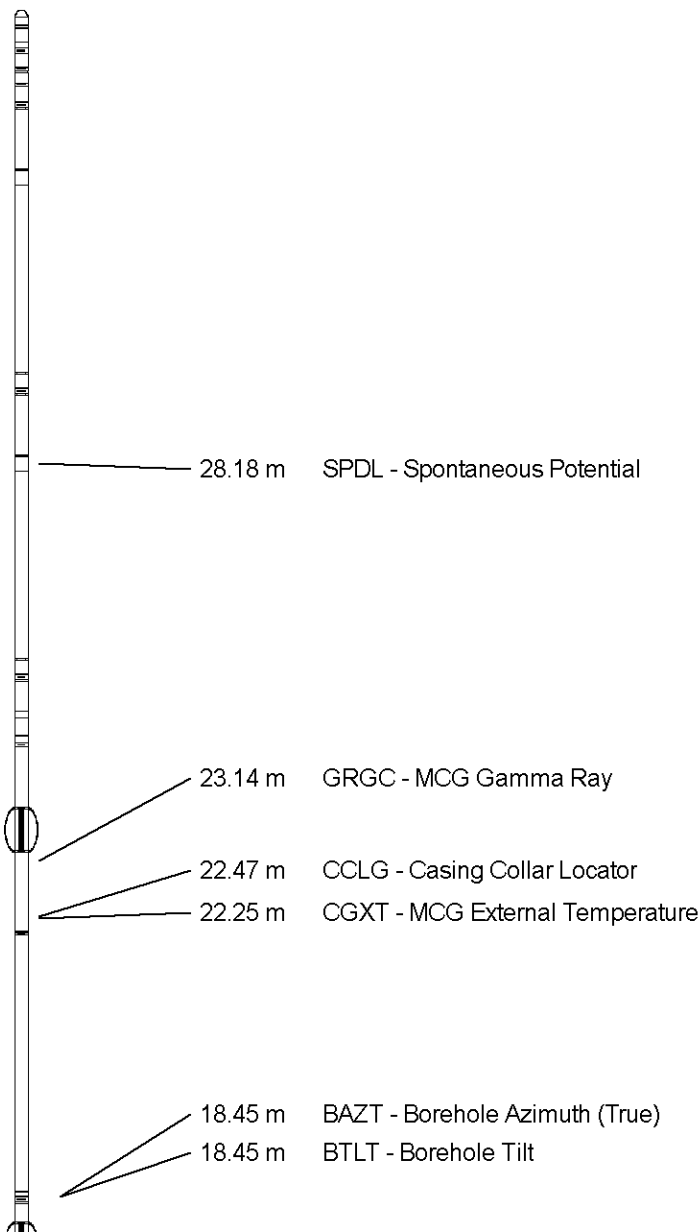
Compact Stiff Bridle Electrode Sub.
MBE-C.B 344 LG: 3.76 m WT: 77.2 lb OD: 2.283 in

Compact Swivel Head Adaptor
SHA-J.B 612 LG: 0.70 m WT: 22.0 lb OD: 2.244 in

Compact Comms Gamma
MCG-E.A 531 LG: 2.65 m WT: 63.9 lb OD: 2.244 in

Compact Navigation
MBN-C.A 55 LG: 3.60 m WT: 70.5 lb OD: 2.244 in

Compact Two Arm Caliper



MTC-B.A 123 LG: 2.17 m WT: 61.7 lb OD: 2.244 in

Compact Sonic
MSS-D.A 446 LG: 3.82 m WT: 72.8 lb OD: 2.244 in

Compact Two Arm Caliper
MTC-B.A 102 LG: 2.17 m WT: 61.7 lb OD: 2.240 in

Compact Knuckle Joint
SKJ-D 31 LG: 0.66 m WT: 24.3 lb OD: 2.244 in

Compact Upper Guard Sub.
MUG-B.A 139 LG: 2.74 m WT: 68.3 lb OD: 2.244 in

Compact Laterolog Electrode Sub.
MLE-D.A 283 LG: 3.76 m WT: 92.6 lb OD: 2.244 in

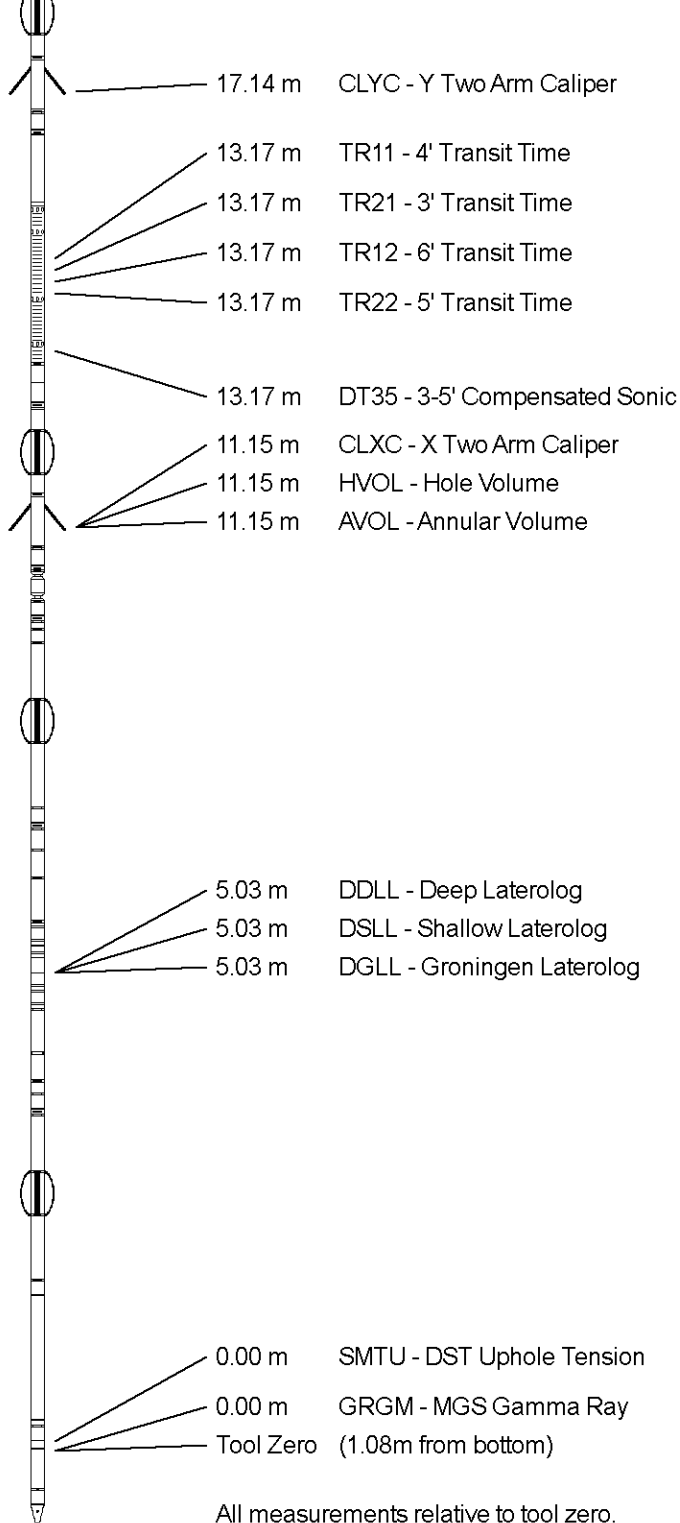
Compact Lower Guard Sub.
MLG-B.A 200 LG: 2.44 m WT: 55.1 lb OD: 2.244 in

Compact Tool Isolator sub.
MTI-B.A 74 LG: 0.47 m WT: 13.2 lb OD: 2.244 in

Compact Short Gamma
MGS-D.A 252 LG: 1.04 m WT: 24.3 lb OD: 2.244 in

Pressure Bung + Hole Finder
HFS 3 LG: 0.28 m WT: 6.6 lb OD: 2.244 in

Total Length: 35.20 m Weight: 828.9 lb



COMPANY	S.N.G.N. ROMGAZ S.A.
WELL	209 NADES
FIELD	NADES
PROVINCE/COUNTY	MURES
COUNTRY/STATE	ROMANIA

Elevation Kelly Bushing	484.71	metres	Last Reading	250.00	metres
Elevation Drill Floor	484.71	metres	First Reading	1212.00	metres
Elevation Ground Level	480.21	metres	Depth Driller	1225.00	metres
			Depth Logger	1226.00	metres



